

Research paper

Hierarchical multi-policy adversarial reinforcement learning

Xiaofeng Wang, Shaopu Wang, Jipeng Zhang^{*,} Yue Ma

Tianjin Key Laboratory for Advanced Mechatronic System Design and Intelligent Control, School of Mechanical Engineering, Tianjin University of Technology, Tianjin, 300384, China

National Demonstration Center for Experimental Mechanical and Electrical Engineering Education, Tianjin University of Technology, Tianjin, 300384, China

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ABSTRACT

In complex and dynamic autonomous driving environments, deep reinforcement learning policies often struggle with generalization due to distribution shifts between training and deployment. While conventional approaches like domain randomization and adversarial training offer partial solutions, they face notable limitations. Specifically, domain randomization relies on manually defined distributions that poorly capture real-world perturbations, while adversarial training suffers from high computational costs, unstable optimization, and behavioral homogenization. To address these issues, we propose a hierarchical multi-policy adversarial reinforcement learning framework that enhances policy generalization under complex perturbations. It incorporates three key components: a multi-policy generation architecture with a shared backbone and independent heads for diverse adversary modeling; a dual-perspective policy expansion regularization mechanism; and an adaptive curriculum-based adversarial training framework. The engineering value translates to a reliable, high-performance solution for autonomous driving, as validated on high-fidelity driving benchmarks. Experimentally, our approach demonstrates superior convergence, stability, and out-of-distribution generalization, achieving performance improvements ranging from 7.36% to 50.06% over established baseline methods while maintaining computational efficiency. These results, obtained in a high-fidelity simulator, demonstrate promise for real-world autonomous driving applications and lay the groundwork for future real-world deployment.

1. Introduction

With the rapid growth of urban traffic and increasing complexity of road environments, autonomous driving has emerged as a critical engineering challenge in intelligent transportation. The core problem lies in developing agents capable of making safe and efficient decisions in highly dynamic and unpredictable traffic conditions. Real-world driving scenarios are inherently uncertain and non-stationary, featuring sudden obstacles, diverse road geometries, and mixed interactions with human drivers. These factors impose severe limitations on conventional rule-based or supervised learning approaches, which often lack the flexibility and adaptability required in realistic driving contexts. In contrast, deep reinforcement learning (DRL) – which enables agents to learn decision-making policies through direct interaction with the environment – offers a promising alternative. DRL provides a unified framework for handling high-dimensional states, continuous control, and long-horizon planning (Zhou, 2024), making it well-suited to address the complex operational demands of future real-world autonomous driving applications.

Over the past decade, DRL has become a cornerstone for solving high-dimensional sequential decision-making problems, yielding remarkable advances across multiple engineering and scientific domains. In autonomous driving and vehicle control, DRL has been applied to end-to-end driving, trajectory planning, and offline policy evaluation. These applications span both robust online learning and benchmark construction for deployment, for example AD4RL (Lee et al., 2024), uncertainty-aware driving (Huang et al., 2024), and multimodal pre-training with RL for autonomous driving (Rowe et al., 2025). Beyond autonomous driving applications, DRL has also demonstrated remarkable success across robotics (Zhao et al., 2025), strategic games (Vinyals et al., 2019b), traffic signal control (Chen et al., 2024), finance (Jiang et al., 2024; Choudhary et al., 2025), and cross-modal decision-making systems (Tobin et al., 2017; Rowe et al., 2025), establishing itself as a versatile framework for complex sequential decision problems.

In safety-critical continuous control tasks like autonomous driving, DRL can learn high-performance policies in simulation. However, it often proves fragile and fails to generalize under real-world challenges

* Correspondence to: Tianjin University of Technology Tianjin, Tianjin, China.

E-mail addresses: wangxiaofeng@tjut.edu.cn (X. Wang), urcwsp@stud.tjut.edu.cn (S. Wang), patriotzjp@email.tjut.edu.cn (J. Zhang), yuema_robot@email.tjut.edu.cn (Y. Ma).

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such as dynamic variations, environmental perturbations, and adversarial behaviors. These limitations can be summarized in three key aspects:

The first major challenge concerns overfitting in adversarial and multi-task training. A common practice is to employ a single adversary, but this leads the protagonist policy to adapt to its fixed perturbation pattern. While yielding strong in-training performance, the policy collapses under out-of-distribution disturbances. Such overfitting restricts generalization, as shown by Li et al. (2021) in parameter-sharing architectures and Yu et al. (2019) in single-task training. Using multiple independent adversaries improves diversity but incurs prohibitive costs. Each adversary requires separate storage and updates, causing computational overhead that scales near-linearly with their number. This makes the approach unsustainable in complex environments where scalability is crucial. The underlying cause is insufficient representational diversity across adversary policies. Single-head networks rely on a single parameterization that cannot capture diverse behavioral patterns, causing policy collapse and instability under distribution shifts (Li et al., 2024). Evolving adversaries further complicate representation in a unified channel, leading to catastrophic forgetting. While multiple independent networks address this, they fragment shared representation learning into redundant training. The resulting sample inefficiency and near-linear cost growth make them impractical for real-world applications requiring both coverage and efficiency.

The second challenge stems from insufficient adversary policy diversity in adversarial training. Limited diversity causes restricted exploration, homogeneous policies that impair cross-task generalization, mismatched challenge levels, and representation collapse across policy heads. As shown in Bettini et al. (2024b), reduced policy diversity in multi-agent systems substantially degrades team performance under sparse rewards. The root cause of insufficient diversity is degeneration of adversary policy distributions and homogenization of gradient updates. When adversaries share reward signals, initializations, and data flows, their update directions become correlated. As a result, action distributions collapse into a low-dimensional subspace, shrinking the range of effective perturbations and causing policy space collapse. Such collapse produces three cascading effects. First, effective sample size reduces as trajectories become redundant, increasing variance in estimates while bias remains high. Second, fragile generalization occurs when the protagonist's learning is restricted to repetitive adversarial modes, causing instability under shifted dynamics. Third, exploration-exploitation imbalance emerges from lacking complementary adversaries. Wang et al. (2021) further confirms that curriculum learning enhances generalization through managed task difficulty and diversity.

The third challenge lies in the inability to reconcile generalization and task completion during training. Single-stage training blends both objectives in one optimization process, where emphasis on stability causes task overfitting while excessive exploration disrupts policy completion. Existing methods also lack hierarchical curricula for varying difficulties, causing easy-task saturation and hard-task underfitting. While diversity constraints offer partial mitigation, their isolation from staged training prevents a dynamic exploration-completion balance. This challenge stems from misaligned training signals and sampling distributions across different training stages and difficulty levels. Simultaneous optimization mixes gradients from different perturbation levels, causing objective conflicts and negative transfer that overfit easy cases and underfit hard ones. Henderson et al. (2018) identified non-stationarity as an instability source where drifting adversary distributions create conflicting gradient directions. Without curriculum staging, value networks suffer systematic misestimation and policy gradients become contaminated. Furthermore, poorly scheduled perturbations yield sparse returns when too strong or cause easy-task saturation when too weak, respectively escalating gradient noise and limiting resilience.

In efforts to improve policy generalization, domain randomization and adversarial training have emerged as two predominant paradigms.

Domain randomization expands policy coverage through random parameter perturbations but relies on manually specified parameter ranges that risk bias against real-world variations. Expanding parameter spaces dilutes samples and reduces efficiency, while narrow ranges omit critical scenarios. As dimensions increase, uniform randomization fails to cover hard-to-learn subspaces and limits out-of-distribution generalization. It also modifies only physical parameters rather than adversarial behaviors, hindering interactive scenario simulation. Chen et al. (2021) found it effective only when randomization matches real variations and policies exhibit memory, while Muratore et al. (2021) showed fixed randomization requires adaptive design and Vuong et al. (2019) emphasized the need for systematic parameter selection. Adversarial training enables direct policy training but demands substantial extra resources and causes instability through concurrent learning. It suffers from convergence bias when adversaries lack diversity, leading to pattern collapse and overfitting to singular perturbations. Furthermore, worst-case training generates zero-reward trajectories that reduce sample utilization and increase gradient noise, with Peng et al. (2021) reporting similar instability issues. Both methods have complementary strengths – spatial coverage for domain randomization and worst-case robustness for adversarial training – yet neither efficiently generates representative diverse samples nor adaptively increases difficulty while maintaining stability. This motivates an integrated approach combining progressive training, hierarchical diversity metrics, and structured multi-adversary modeling to enhance generalization in complex driving environments.

To address the above challenges, we propose an adaptive curriculum-based adversarial training framework designed for complex dynamic environments. This framework integrates a multi-policy generation architecture with a multi-view policy diversification regularizer, enabling robust generalization of the protagonist policy under out-of-distribution dynamics. Specifically:

First, we introduce a multi-policy generation architecture (MPG-Arch) combined with multi-environment parallelization to enhance policy adaptability and generalization in complex adversarial scenarios. Multiple policy heads provide independent parameterization channels, each specializing in a distinct behavioral mode, while dedicated value heads ensure localized and accurate advantage estimation. The shared backbone extracts low-level features, while lightweight multi-heads avoid the prohibitive costs of maintaining fully independent networks. Concurrently, multiple instances of the same environment host diverse adversaries simultaneously, efficiently fusing heterogeneous experiences to stabilize value estimation and accelerate policy iteration. Together, these designs enhance diversity and coverage of adversarial perturbations while ensuring computational scalability and efficiency.

Second, we employ a dual-perspective policy expansion regularization (DPPER) that dynamically adjusts adversary policy distributions, encouraging adversaries to explore broader and more challenging behavioral patterns to increase distributional coverage. The hierarchical covariance diversity principle maximizes both within-state and across-state action covariance volume, generating explicit, multi-directional gradient signals at both statistical and optimization levels. This prevents the collapse of adversary policies into low-rank or highly correlated modes, ensuring persistent heterogeneity in adversarial behavior patterns. It expands explorable perturbation subspaces, reduces sample redundancy, and addresses core challenges in adversarial training including “effective sample size reduction, fragile generalization support, exploration–exploitation imbalance, and selection bias”. The overall design directly targets conventional method limitations including policy overfitting, homogeneous adversarial behavior, and insufficient generalization, fundamentally enhancing the protagonist policy's ability to handle unknown perturbations and dynamic interactions.

Third, we propose an adaptive curriculum-based adversarial training framework (ACATF). It decouples optimization of generalization objectives into three sequential stages – (I) foundational skill acquisition, (II) distribution expansion, and (III) adversarial enhancement – rather

than blending heterogeneous gradients within a single update. This avoids a negative transfer from simultaneously mixing perturbations with different difficulties. In Stage II, perturbation statistics are fixed and rewards/advantages normalized, while variance-sensitive hyperparameters of GAE are adjusted to stabilize value learning and mitigate bias across dynamics. Adversary strength escalates adaptively based on performance thresholds, preventing zero-reward trajectories. Each stage maintains near-stationary perturbation statistics, with controlled transitions only at stage boundaries.

The main contributions of this paper are summarized as follows:

- A computationally efficient multi-policy generation architecture (MPG-Arch) that enables parallel generation of diverse adversarial policies with low overhead. Integrated with parallelized environments, it supports scalable distributed sampling and optimization, significantly improving both perturbation coverage and training throughput—key factors for deploying robust driving policies in practice.
- A geometrically interpretable diversity regularization mechanism (DPPER) that maintains adversarial heterogeneity from both within-state and across-state perspectives. By preventing mode collapse and ensuring a broad spectrum of challenges, it enhances policy robustness in open-world driving scenarios where environmental conditions vary continuously.
- An adaptive curriculum-based adversarial training framework (ACATF) that structures training into staged phases—from skill acquisition to randomized dynamics and finally targeted adversarial training. This approach decouples learning objectives and perturbation profiles, leading to more stable convergence and stronger generalization under distribution shifts, as validated in high-fidelity driving environments.

The remainder of this paper is structured as follows: Section 2 reviews relevant prior work in the field; Section 3 presents our core methodology; Section 4 details all conducted experiments and results; and Section 5 provides a comprehensive summary.

2. Related work

2.1. Robust reinforcement learning

In recent years, the uncertainty of environment and the prevalence of adversarial perturbations in complex domains such as autonomous driving, robotic control, and multi-agent games have significantly challenged the operational stability and cross-scenario generalization of reinforcement learning policies. As a result, robust reinforcement learning has gradually emerged as a key and urgent research direction in the field.

One mainstream approach is domain randomization (DR), whose core idea is to randomly sample environment parameters (e.g., physical dynamics, sensor noise, or perturbation magnitudes) during training to encourage policies to learn across broader distributions and acquire cross-domain adaptability (Tobin et al., 2017). DR has been widely applied in robotic manipulation and autonomous driving simulations, demonstrating substantial success in sim-to-real transfer. However, its reliance on manually specified parameter distributions often makes it difficult to cover rare but critical perturbation patterns in real-world environments, resulting in failure risks in out-of-distribution settings.

Another important approach is adversarial reinforcement learning. A typical representative is robust adversarial reinforcement learning (Pinto et al., 2017), which introduces an adversarial agent that actively learns to generate targeted perturbations, forcing the protagonist agent to train under worst-case environmental conditions and thus significantly improving its overall robustness. Subsequent studies have further extended this idea with diverse adversarial modeling (Pan et al., 2019), game-theoretic optimization (Zhang et al., 2020), and

distributionally robust methods (Mehta et al., 2020), which collectively enhance policy stability and generalization under complex perturbation conditions. Furthermore, Wang et al. (2024) proposed an adversarial-perturbation-based deep RL framework that achieved robustness across multiple high-dimensional dynamic environments.

Despite these advances, adversarial training still suffers from several inherent and persistent limitations. Adversarial games often result in large computational overhead and unstable convergence. Moreover, if adversaries are too strong, protagonist learning may stagnate; if adversarial policies converge to overly homogeneous patterns, the protagonist tends to overfit and exhibits poor out-of-distribution generalization. Therefore, achieving an effective balance between efficiency, stability, and diversity remains a critical challenge in robust reinforcement learning.

2.2. Multi-head neural networks

In recent years, multi-head architectures have emerged as an important structural innovation in deep learning, primarily due to their ability to achieve multi-task decoupling and multi-modal modeling based on a shared feature backbone. This design has been widely adopted across diverse domains.

In computer vision, multi-head structures were first extensively studied in object detection tasks. For instance, Faster R-CNN (Ren et al., 2015) and Mask R-CNN (He et al., 2017) both adopt shared low-level perception backbones with multiple independent classification and regression heads at higher layers, thereby enabling decoupled modeling of object categories and scales. Such designs have since been widely extended to instance segmentation and keypoint detection tasks (Cao et al., 2017; Hestness et al., 2019). In multi-task learning, multi-head architectures allow for the joint optimization of related yet heterogeneous tasks—for example, object recognition with attribute prediction (Caruana, 1997) or the combined modeling of geometric and semantic properties (Kendall et al., 2018).

More recently, multi-head attention mechanisms have become the central component driving breakthroughs in transformer-based models, and their effectiveness has been widely validated in vision and control domains. In computer vision, multi-head attention and its variants have been successfully employed to construct highly effective and generalizable backbones. For example, the swin transformer introduced a hierarchical attention architecture with shifted windows, achieving state-of-the-art performance in image classification and object detection (Liu et al., 2021). Subsequent models such as cross-shaped window transformer and other multi-scale transformer variants further refined head and window designs, significantly improving multi-scale representation learning and advancing performance in classification, detection, and video understanding tasks (Dong et al., 2022; Jiao et al., 2023).

In natural language processing, multi-head attention has remained the core mechanism for sequence modeling since the original transformer. Recent studies have examined the role and redundancy of attention heads from both theoretical and empirical perspectives, proposing grouping and selection mechanisms to improve head utilization and model generalization (Ni et al., 2023). Simultaneously, efforts to extend multi-head structures toward greater modularity and sparsity have gained attention in the machine learning community. Wu et al. (2024) treat “head” as an expert and introduce a multi-head design in the mixture-of-experts style, which effectively improves representation density and computational efficiency.

These studies collectively show that carefully designed multi-head structures can significantly improve model representation capacity and generalization while maintaining computational efficiency, providing both an engineering foundation and theoretical motivation for extending multi-head mechanisms to reinforcement learning and adversarial training.

2.3. Diversity

In adversarial training and multi-agent reinforcement learning, introducing policy diversity has been widely recognized as a key mechanism to enhance generalization. Existing approaches typically incorporate diversity regularization terms into the optimization objective, constraining the dissimilarity among policies to prevent mode collapse of adversaries or sub-policies.

A representative approach leverages cosine similarity constraints, which measure directional differences between action vectors of different policies, thereby encouraging orthogonality or low correlation in the action space (Eysenbach et al., 2018). This method is computationally efficient and geometrically interpretable, effectively preventing multiple heads from producing nearly identical actions under the same state. However, it neglects magnitude information and often fails in high-dimensional continuous action spaces. Another common approach is based on Kullback–Leibler (KL) divergence, which evaluates the relative entropy between policy distributions to enforce diversity at the probabilistic level (Haarnoja et al., 2018). While KL divergence provides a finer characterization of distributional differences, it is computationally expensive and prone to gradient vanishing or numerical instability when distributions overlap minimally. Moreover, in multi-head or multi-agent scenarios, KL divergence typically requires pairwise computation, causing quadratic growth in cost as the number of policies increases, thereby limiting scalability in large-scale adversarial settings.

Recent studies have further expanded the scope of diversity modeling. For example, Tang et al. introduced the policy similarity measure (Tang et al., 2025), which combines Gaussian distance and cosine similarity to evaluate inter-policy similarity and incorporates this metric as a regularizer in the policy-space response oracles framework, effectively reducing exploitability and enhancing diversity in environments such as Leduc Poker. In recommendation systems, Wang et al. proposed the diversity-enhanced counterfactual recommendation model (Wang et al., 2025), which employs feature entropy and dynamic reweighting of long-tail data to improve recommendation policy diversity while implicitly balancing KL/entropy/similarity conflicts, thereby mitigating popularity bias and recommendation homogeneity. In language modeling and reasoning, researchers have further revealed the critical role of divergence selection. Long Li et al. (2025) demonstrated that forward KL, reverse KL, and Jensen–Shannon divergence significantly impact policy diversity and knowledge retention, emphasizing the use of mass-covering divergences to resist the “policy compression” effect of reverse KL. Similarly, VendiRL (Lintunen, 2025) introduced the Vendi Score, a flexible metric that accommodates arbitrary similarity functions (e.g., cosine distance), thus enabling more adaptable diversity modeling in high-dimensional policy spaces.

Beyond distribution-level constraints, some works have explored behavioral and skill-level diversity. Bettini et al. (2024a) proposed an adaptive reinforcement learning method that explicitly regulates agent behavioral diversity, effectively alleviating convergence toward homogeneous policies in multi-agent settings. Qing and Zhu (2024) introduced a clustering-based skill diversity framework, reducing redundant exploration and improving representational richness and generalization. Xin et al. (2025) further applied vectorized diversity to path planning, enabling policies to acquire good transferable capabilities within a short training horizon.

2.4. Curriculum learning

Curriculum Learning was first introduced by Bengio et al. (2009), with the central idea of presenting tasks in a progressively increasing order of difficulty to improve convergence speed and generalization ability. In reinforcement learning, this paradigm has been widely introduced into adversarial training to mitigate training instability and task difficulty imbalance. Existing approaches primarily fall into two

categories. The first is environment curricula, where the complexity of the environment or the strength of perturbations is gradually increased, enabling the policy to acquire robustness along a controlled learning trajectory. For example, Chen et al. (2021) progressively expand the sampling range of environment parameters, thereby improving the generalization of policies under out-of-distribution conditions. The second is adversary curricula, where the strength or diversity of adversaries is dynamically adjusted, forcing the protagonist to gradually adapt to increasingly challenging disturbances. To address this issue, Gur et al. (2021) proposed an adaptive difficulty design, enabling adversaries to generate moderately challenging disturbances, which effectively enhances policy robustness and generalization.

Recent work has integrated curriculum learning with adversary modeling and diversity constraints to further address generalization challenges from environmental perturbations and task complexity. For instance, Satheesh et al. (2025) introduced a curriculum framework guided by contextual variables and language models, where a diversity-based context blending mechanism is incorporated to prevent disturbance mode collapse and improve adaptability in heterogeneous environments. Similarly, PPO-ACT (Yang et al., 2025) adopts adversarial curriculum transfer in public goods games, allowing the policy to gradually improve robustness through two-stage training with weak-to-strong adversaries, demonstrating the potential of combining curriculum and adversarial approaches. Another line of work, trajectory first (Braun et al., 2025), emphasizes trajectory-level exploration of diversity before step-level fine-tuning, achieving higher policy diversity and robustness in challenging robot control tasks. In multi-agent settings, counterfactual group relative policy advantage (Jin et al., 2025) introduces counterfactual group relative policy advantage to dynamically adjust task difficulty and adversary strength, thereby stabilizing curriculum training and reducing interference among adversaries.

In large-scale multi-agent games, curriculum-based adversarial training has become a cornerstone paradigm. A representative example is AlphaStar (Vinyals et al., 2019a), which employs the League Training mechanism to organize protagonists, adversaries, and explorers into a dynamic league. This framework enables a curriculum-like progression of difficulty, effectively avoiding convergence to a single adversary mode and ensuring sustained progress in diversity and robustness. A similar core policy is adopted in OpenAI Five (Berner et al., 2019), where iterative self-play is combined with curriculum-based adversarial dynamics, allowing agents to gradually adapt to increasingly complex in-game environments over time.

3. Method

Fig. 1 provides a systematic overview of the proposed algorithm framework and its core workflow. We first construct a group of N parallel environment instances, denoted as $\{\text{Env}_i\}_{i=1}^N$. From the environment group, we obtain the state group $S_t = \{s_{i,t}\}_{i=1}^N$ and the reward group $R_t = \{r_{i,t}\}_{i=1}^N$ at timestep t . The subscript i in $s_{i,t}$ and $r_{i,t}$ denotes the environment index.

The reward group R_t is used to update the parameters of both the protagonist and the adversary group ADV . As for the state group S_t , it is received simultaneously by the protagonist and ADV , each of which generates its own corresponding action group.

The protagonist generates actions using a standard single-network (i.e., a single shared backbone and a single policy head), producing the protagonist action group $\{a_{i,t}^p\}_{i=1}^N$ for the N parallel environment instances. Concurrently, the adversary group ADV , equipped with the MPG-Arch module, processes the same S_t . The MPG-Arch employs a shared backbone for feature extraction and N independent policy heads, denoted as $adv_1, adv_2, \dots, adv_N$, to generate distinct adversarial policies in parallel. Each policy head j , denoted as adv_j , generates its own action group, which consists of $\{a_{i,t}^{adv_j}\}_{i=1}^N$. The ADV action group $\{a_{i,t}^{adv}\}_{i=1}^N$ is formed by selecting, for each environment instance Env_i , the action from the adversary with the matching index i .

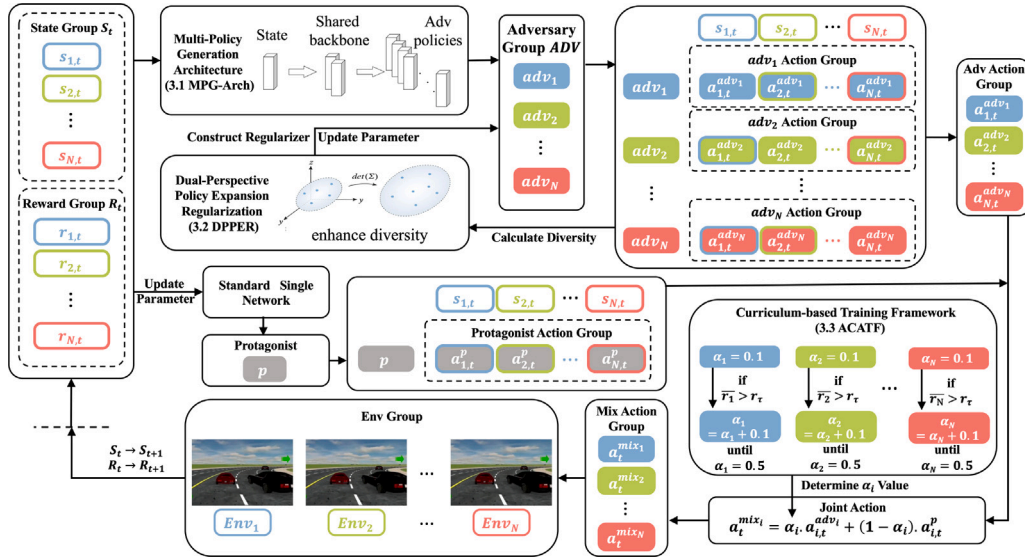


Fig. 1. Overall overview of the proposed algorithm.

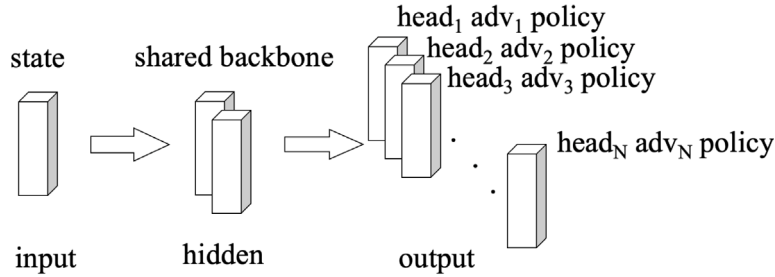


Fig. 2. Schematic diagram of the multi-policy generation architecture.

During the action mixing stage, $a_{i,t}^p$ is mixed with $a_{i,t}^{adv_i}$ to produce $a_t^{mix_i}$, forming mix action group $\{a_t^{mix_i}\}_{i=1}^N$. Applying this mix action group to the environment group (i.e., action $a_{i,t}^{adv_i}$ is applied to Env_i) yields the next state group S_{t+1} , whilst simultaneously obtaining the reward group R_{t+1} .

Within the dual-perspective policy expansion regularization (DP-Per) module, within-state and across-state regularizations are computed (see Section 3.2 for details). The resulting regularization term multiplied by the diversity coefficient, is used to diversify adversary policies, thus improving the generalization of the protagonist.

Finally, the adaptive curriculum-based adversarial training framework (ACATF) dynamically adjusts adversarial strength during training by progressively increasing opponent difficulty and curriculum complexity, thereby guiding the protagonist toward enhanced generalization capability. Implementation details are provided in Section 3.3.

3.1. Multi-policy generation architecture (MPG-Arch)

To enhance both policy diversity and training efficiency, we design a shared-backbone multi-policy generation architecture (MPG-Arch) as the core architecture of the proposed adversarial reinforcement learning framework. Built upon a unified backbone neural network, MPG-Arch branches into multiple policy output heads, each representing a distinct adversary with heterogeneous behavioral patterns. Together, these heads form a diverse adversarial policy ensemble (see Fig. 2 for an illustration of the multi-head architecture). By combining structural sharing with head-level separation, MPG-Arch enables efficient learning of multiple adversary policies, making it particularly well suited for autonomous driving environments. In contrast, the

protagonist adopts a standard single-policy network, since the goal is to learn one unified and robust driving policy for deployment rather than maintaining a diverse set of behaviors.

The shared-backbone design of MPG-Arch is also supported by classical multi-task learning (MTL) theory. Caruana (1997) argues that learning multiple related tasks jointly promotes information sharing across tasks, encouraging the model to reuse representations that are predictive across objectives and thereby improving sample efficiency and reducing overfitting compared to training each task independently, while Baxter (2000) provides a learning-theoretic justification by formalizing MTL as learning a shared representation that induces a beneficial inductive bias—constraining multiple tasks to rely on a common feature extractor effectively restricts the hypothesis space toward representations that generalize better. In our setting, each adversary head can be viewed as a distinct but related “task” (a different adversarial behavior mode) operating on the same driving state space, so the shared backbone enables principled transfer of representation knowledge across heads while still allowing head-specific specialization. Moreover, aggregating gradients from multiple adversary heads when updating the shared backbone provides an additional regularization effect: because different heads emphasize different regions of the state-action space and induce different optimization signals, their gradient aggregation discourages over-specializing to any single adversarial mode and tends to smooth training dynamics, resembling an implicit ensemble effect—while the heads maintain diverse policies, the shared representation is optimized to be broadly useful under multiple behavioral objectives, analogous to how ensembles improve generalization by combining diverse predictors.

In this design, the shared backbone ensures efficient representation reuse and stable feature extraction across different adversaries,

while the independent heads encourage behavioral differentiation and reduce policy redundancy. This design enables MPG-Arch to achieve strong scalability and adaptability in complex, high-dimensional, and dynamically changing environments.

The architecture consists of two key modules: (1) a shared feature extraction backbone, and (2) multiple policy-value output heads. The shared backbone, implemented as a multi-layer network, encodes the state into a unified high-dimensional representation that is shared across all policy heads. This design significantly reduces parameter redundancy and computational overhead during forward inference while avoiding repeated learning of redundant state features. It also enhances training efficiency and stability by providing a consistent representational foundation for downstream policies.

Built upon the shared representation, each policy head consists of an independent adversary policy generator $\pi_i(a|s)$ and a corresponding value estimator $V_i(s)$. Given the same state input, these generators produce stylistically diverse action distributions, thereby structurally enforcing behavioral heterogeneity and broadening the exploration of the policy space. This design effectively alleviates policy collapse (where multiple policies converge to similar behavioral patterns), reduces the risk of local optima, and promotes parallel exploration across distinct subspaces of the policy space. Importantly, the outputs of different heads are never aggregated or averaged; each head adv_i is assigned exclusively to environment instance Env_i , contributing an independent and distinct perturbation stream to the training process.

From a computational perspective, the MPG-Arch provides substantial benefits over independent multi-network designs. By sharing the feature extraction backbone, the MPG-Arch substantially reduces the total parameter count, memory consumption, and inference latency—making it especially suitable for deployment on resource-constrained edge devices such as automotive system-on-chips and embedded GPUs.

Moreover, the MPG-Arch employs a unified gradient update mechanism. The parameters of policy head i , denoted as θ_i for $i \in (1, \dots, N)$, and the parameters of the shared backbone, θ_b , are updated according to the following rules, respectively:

$$\theta_i \leftarrow \theta_i - \alpha \nabla_{\theta_i} (\mathcal{L}_i^{DRL} + \lambda REG), \quad (1)$$

$$\theta_b \leftarrow \theta_b - \alpha \nabla_{\theta_b} \left(\sum_{i=1}^N [\mathcal{L}_i^{DRL} + \lambda REG] \right), \quad (2)$$

where α is the learning rate, $\mathcal{L}_i^{DRL}$ is the deep reinforcement learning objective loss for the i th head, REG is the globally computed regularization term (detailed in Section 3.2), and λ is the regularization coefficient.

The update for θ_i specializes the i th adversarial policy, enhancing its perturbation efficacy while conforming to its specific regularization constraints. Since each head is updated independently, this process fosters behavioral diversity across the adversary group. In contrast, the update for θ_b integrates gradient signals from all N adversary heads. This encourages the backbone to function as a “generalist”, learning feature representations that simultaneously support the success of all diverse adversary policies. Consequently, the backbone is compelled to discard features that are overspecialized to any single policy—which may lead to overfitting—in favor of acquiring more robust and generalizable environmental representations.

Furthermore, MPG-Arch offers significant scalability and deployment flexibility, as the number of policy heads can be readily adjusted to match specific task complexity and diversity requirements.

In summary, the MPG-Arch integrates efficient feature sharing with inherent policy diversity within a unified adversarial learning framework, providing a scalable and computationally efficient foundation for multi-policy learning in complex autonomous driving tasks.

3.2. Dual-perspective policy expansion regularization (DPPER)

To further enhance the behavioral diversity among adversaries, we propose a novel and comprehensive dual-perspective policy expansion regularization (DPPER) mechanism (see Fig. 3). Drawing inspiration from hierarchical analysis, this mechanism models the structural differences between adversary policies at two complementary levels: a within-state perspective and an across-state perspective, imposing a diversity-driven regularizer to strengthen the exploratory capacity and stability of the overall adversarial policy ensemble.

The within-state perspective quantifies the behavioral diversity across all adversaries conditioned on the same state s_t . Given the same state s_t , the N adversaries yield a set of actions $\{a_t^{adv_1}, a_t^{adv_2}, \dots, a_t^{adv_N}\}$, forming an adversary action matrix $\mathbf{J}_1 \in \mathbb{R}^{N \times d}$, where d is the action dimension. The determinant of its covariance matrix, $\det(\Sigma_{\mathbf{J}_1})$, thus measures the dispersion of adversary responses in the action space. A larger value indicates greater behavioral diversity under the same state.

The across-state perspective captures the global behavioral structure of adversary policies across states over multiple time steps. Specifically, the adversary actions from past $M = \min(m, t)$ time steps (from $t - M + 1$ to t) are concatenated, where m is a fixed window size, to form a joint adversary action matrix $\mathbf{J}_2 \in \mathbb{R}^{N \times (M \cdot d)}$, from which the covariance matrix $\Sigma_{\mathbf{J}_2}$ is computed. The determinant $\det(\Sigma_{\mathbf{J}_2})$ reflects the overall breadth and coverage of adversary behaviors across the states in the window, thereby quantifying long-horizon diversity that extends beyond single-state variability.

To combine both perspectives, we define the final hierarchical diversity indicator as:

$$REG = \log \det(\Sigma_{\mathbf{J}_1}) + \frac{1}{M} \sum_{i=1}^M \log \det(\Sigma_{\mathbf{J}_2}) \quad (3)$$

where M is the number of sampled states. Geometrically, the determinant of a covariance matrix corresponds to the volume of the ellipsoid spanned by the action distribution, where a larger determinant indicates more dispersed policy outputs and thus provides a clear and interpretable measure of diversity. In the proposed method, the final diversity indicator is obtained by combining the quantification results from both within-state and across-state levels in a weighted manner, which not only preserves fine-grained discrimination among individual adversaries but also enhances the overall differentiation across multiple adversarial policies.

From a mathematical standpoint, our determinant-based hierarchical regularizer differs fundamentally from spectral-entropy diversity metrics such as the Vendi Score. DPPER explicitly maximizes $\log \det(\Sigma)$, where Σ is the covariance of the adversary action set (within-state) or the concatenated action trajectory (across-state). Since $\det(\Sigma) = \prod_{i=1}^d \lambda_i(\Sigma)$, the objective directly encourages all principal variances to expand, i.e., it increases the volume of the ellipsoid spanned by adversary actions in the continuous control space, yielding a clear geometric interpretation and a stable, differentiable training signal. In contrast, the Vendi Score is defined via the exponentiated entropy of the eigenvalue spectrum of a normalized kernel matrix $\rho = K/\text{tr}(K)$, i.e., $\text{VS} = \exp(-\sum_i \lambda_i(\rho) \log \lambda_i(\rho))$, where $K_{ij} = k(x_i, x_j)$ depends on a user-specified similarity function $k(\cdot, \cdot)$. Because the similarity function must be manually specified (and tuned in scale), the induced notion of diversity can be sensitive to this choice and may not be universally well-matched to continuous-control behaviors in autonomous driving. As a result, its notion of diversity is implicitly determined by kernel choice and scale, and computing VS typically requires constructing an $O(N^2)$ kernel matrix and performing spectral decomposition, which is substantially more expensive than covariance estimation in our low-dimensional action space. In comparison, DPPER operates directly on already-sampled actions, with a per-state covariance computation of $O(Nd^2)$ (or $O(Nd)$ for diagonal covariance), and thus scales more favorably for multi-head adversarial RL where diversity must be evaluated repeatedly during training. This difference makes DPPER particularly

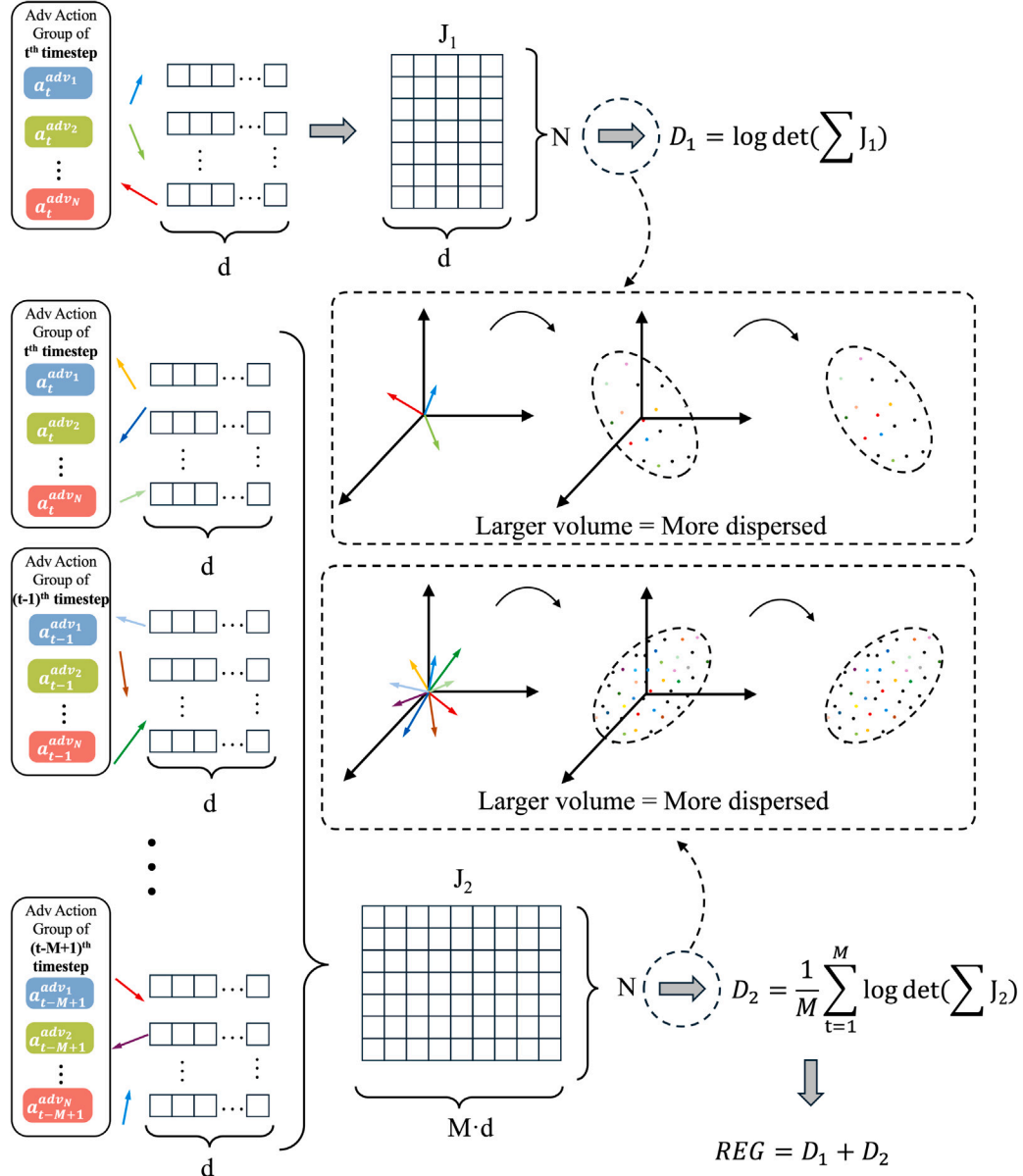


Fig. 3. Illustration of the dual-perspective policy expansion regularization.

suitable for autonomous driving, where diversity regularization should be both geometrically meaningful and computationally lightweight.

More importantly, this hierarchical regularization design offers strong geometric interpretability and naturally integrates into reinforcement learning loss functions as a unified and optimizable training objective. Unlike simple pairwise Euclidean distances that can be dominated by action magnitude, DPPER measures diversity through the covariance structure of (normalized) action outputs, where the log-determinant captures the multi-dimensional spread and coupling among action dimensions. This encourages adversaries to diversify along multiple principal directions and correlation patterns in the action space, rather than merely increasing the amplitude of individual control signals. Its hierarchical structure enables multi-level analysis of policy dispersion, while its interpretability and compatibility make it particularly well-suited for autonomous driving tasks that require real-time responsiveness, safety-aware decision-making, and stable learning in high-dimensional action spaces.

The proposed DPPER demonstrates significant advantages in autonomous driving scenarios. By jointly modeling adversary policy diversity from both within-state and across-state perspectives, it effectively mitigates common issues such as adversary policy homogenization and behavioral convergence, thereby enhancing the generalization capability of the protagonist policy. The incorporation of diverse adversarial behaviors ensures richer and more representative training interactions, enabling the protagonist to develop sufficient adaptability across varied and unseen driving conditions. Specifically, the state-wise modeling captures the instantaneous diversity of adversary actions at each timestep, while the joint-level modeling characterizes the long-term trajectory coverage across sequences of states, ensuring that adversarial perturbations remain both dynamic and informative.

Compared to a single global diversity metric (cosine, vendi score) computed over all samples, our hierarchical design provides a more targeted and stable learning signal. The within-state term explicitly enforces that different heads respond differently under the same driving

Table 1
Stage transition and termination criteria.

Stage transition	Criterion
Stage 1 → Stage 2	Mean reward over last 20 episodes $> \tau$
Stage 2 → Stage 3	Mean reward over last 20 episodes $> \tau$
Stage 3 termination	All $\alpha_i \geq 0.5$ or epoch ≥ 60000

context, preventing trivial diversity that only arises from averaging over heterogeneous states. Meanwhile, the across-state term encourages each adversary to maintain broad behavioral coverage over temporally correlated state sequences, reducing mode collapse over long horizons and discouraging policies that are diverse only at isolated timesteps. This decomposition therefore disentangles instantaneous action dispersion from long-horizon behavioral coverage, leading to stronger discrimination among adversaries, improved robustness under distribution shift, and more consistent adversarial pressure during training.

Moreover, this mechanism introduces almost no additional computational burden, as it relies solely on covariance estimation of already-sampled action data. The determinant-based computation is differentiable and can be efficiently integrated into the existing training loop, avoiding extra forward passes or gradient evaluations, thus maintaining the overall efficiency of the reinforcement learning process.

In summary, the proposed DPPER regularization mechanism effectively enhances the behavioral coverage, differentiation, and stability of adversary policies without significantly increasing computational burden. It provides a principled and effective foundation for deploying multi-head architectures in autonomous driving decision systems.

3.3. Adaptive curriculum-based adversarial training framework (ACATF)

To comprehensively enhance the generalization of the protagonist agent in complex and dynamic environments, we propose an adaptive curriculum-based adversarial training framework (ACATF). This framework integrates curriculum learning with perturbation augmentation under a progressive staging principle. Following a simple-to-complex progression paradigm, the training process is divided into three stages – default parameter training, randomized perturbation training, and adversarial perturbation training – progressively increasing task complexity and disturbance intensity. This staged design guides the protagonist toward a balanced trade-off between stability and adaptability, ultimately producing high-quality, disturbance-resistant policies. The specific switching criteria are shown in Table 1.

Stage I: Default Environment Training.

In this stage, the environment dynamics (e.g., vehicle mass, tire-road friction coefficient, steering sensitivity) are fixed to default values, with minimal external disturbances. This phase enables the protagonist to rapidly master basic task capabilities such as trajectory following, obstacle avoidance, and speed regulation, thereby establishing a complete perception–decision–execution pipeline. The stable environment ensures high convergence of policy optimization and lays a solid foundation for subsequent training stages.

Stage II: Randomized Perturbation Training.

In this stage, environment dynamics are randomly sampled at the beginning of each episode to replicate real-world uncertainties such as tire wear, payload variations, or diverse road conditions. The protagonist must accomplish tasks under unstable dynamics, substantially improving adaptability to system variations and out-of-distribution generalization. Furthermore, this increased diversity in training distributions mitigates the risk of policy overfitting.

Stage III: Adversarial Perturbation Training.

In the final stage, heterogeneous adversary policies are introduced. Rather than altering environment parameters directly, these adversaries perturb the protagonist's control. This is formalized by blending the adv

action group with the protagonist action group using a linear mixing mechanism:

$$a_t^{\text{mix}_i} = \alpha_i \cdot a_{i,t}^{\text{adv}_i} + (1 - \alpha_i) \cdot a_{i,t}^p \quad (4)$$

where $a_{i,t}^p$ is the i th action of the protagonist action group, $a_{i,t}^{\text{adv}_i}$ is the i th action of the adv action group, and $\alpha_i \in [0, 1]$ is mixing coefficient. This formulation establishes a continuously tunable game that requires the protagonist to maintain task performance while resisting adversarial perturbations.

To achieve adaptive adjustment of the confrontation intensity, this paper proposes a mixed coefficient adjustment mechanism based on performance feedback. Initially, each adversary's blending coefficient α_i is set to 0.1, granting the protagonist dominant control in decision-making. Every fixed interval (e.g., 20 episodes), the average episodic reward $\bar{r}_i$ against adversary i is evaluated. If $\bar{r}_i$ exceeds a predefined threshold τ (set τ to 1200 in our experiments), α_i is increased by 0.1, thus raising the influence of i th adversary. The value of α_i is capped at 0.5. If the protagonist continues to perform well even under maximum adversarial strength, the corresponding adversary is marked as defeated and removed from the training pool.

The complete Stage III procedure, including initialization, per-episode updates, and adversary deactivation logic, is summarized in Algorithm 1.

Algorithm 1 Stage-III Adversarial Training with Action Mixing

Require: Protagonist p , adversary set $ADV = \{adv_i\}_{i=1}^N$, environment $Env = \{Env_i\}_{i=1}^N$
Require: Max epochs E_3 , initial mixing α_0 , reward threshold τ , increment $\Delta\alpha$, $\alpha_{\max}$, check interval K_α
Ensure: Updated protagonist p , updated adversaries ADV
1: Initialize $\alpha_i \leftarrow \alpha_0$, $\text{active}_i \leftarrow \text{True} \quad \forall i \in \{1, \dots, N\}$
2: **for** epoch = 1, 2, ..., E_3 **do**
3: **if** $\nexists i : \text{active}_i = \text{True}$ **then**
4: **break**
5: **end if**
6: **for each** adversary $adv_i \in ADV$ **do**
7: **if** $\text{active}_i = \text{False}$ **then**
8: **continue**
9: **end if**
10: Reset $Env_i \rightarrow s$
11: Run one episode with mixed action $a_{\text{mix}} = \alpha_i a_{adv_i} + (1 - \alpha_i) a_p$
12: Collect protagonist return R_p
13: Update protagonist p using collected trajectories
14: Set adversary return: $R_{adv_i} \leftarrow -R_p$
15: Update adversary adv_i
16: **end for**
17: **if** epoch mod $K_\alpha = 0$ **then**
18: Evaluate protagonist only (no adversary active)
19: **for** $i = 1, \dots, N$ **do**
20: **if** active_i **and** $\text{mean_recent}(R_{adv_i}) < -\tau$ **then**
21: $\alpha_i \leftarrow \alpha_i + \Delta\alpha$
22: **end if**
23: **if** $\alpha_i \geq \alpha_{\max}$ **then**
24: $\text{active}_i \leftarrow \text{False}$
25: **end if**
26: **end for**
27: **end if**
28: **end for**

From a principled perspective, the unified transition threshold τ is calibrated based on the protagonist's stabilized reward after Stage I and Stage II convergence, set to approximately 80%–90% of that reference competence level. Although τ is the exact same scalar for both transitions, it plays distinct roles at each curriculum boundary. In Stage I, the environment is deterministic and disturbance-free, allowing the agent to consistently achieve stable returns; thus, τ serves as an easily attainable threshold that ensures basic driving competency is acquired

before moving forward. Conversely, for the transition from Stage II to Stage III, this identical τ becomes a rigorous standard. It guarantees that the policy maintains high-quality performance under randomized dynamics before facing adversarial perturbations. Overall, this unified design ensures that adversarial intensity is strengthened only after the policy reaches a sufficiently strong baseline performance, successfully avoiding premature escalation or overly lenient progression in Stage III. Moreover, the upper bound $\alpha_i \leq 0.5$ is imposed to prevent the adversary from becoming dominant in the mixed action. When α_i approaches 0.5, the adversary's control authority in $a^{\text{mix}} = \alpha_i a^{\text{adv}_i} + (1 - \alpha_i) a^p$ becomes comparable to the protagonist; if α_i were allowed to exceed 0.5, the adversary would effectively override the protagonist and drive the executed control toward adversarial intent, which can lead to catastrophic behaviors and destabilize learning. Therefore, setting the cap at 0.5 enforces a “strong-but-bounded” disturbance regime that remains challenging while still allowing the protagonist to retain sufficient control for meaningful policy improvement.

The above action-mixing mechanism naturally induces an adversarial game between the protagonist and the adversary. Concretely, the protagonist policy $\pi_p(\cdot | s; \theta_p)$ aims to maximize the task return, while each adversary head $\pi_{\text{adv}_i}(\cdot | s; \theta_{\text{adv}_i})$ is optimized to maximally degrade the protagonist's performance (i.e., using the negated protagonist return as the adversary objective). This yields an approximately zero-sum formulation at the objective level. However, because the training follows a curriculum-style progressive procedure with performance-driven intensity scheduling, the practical optimization is closer to a leader–follower (Stackelberg-style) process, where the protagonist is updated continuously and each adversary adapts as a follower to the current protagonist under the scheduled mixing coefficient α_i .

Formally, let $s_{i,t}$ denote the state of environment instance Env_i at time t , and the executed action be $a_{i,t}^{\text{mix}} = \alpha_i a_{i,t}^{\text{adv}_i} + (1 - \alpha_i) a_{i,t}^p$ as defined above. Define the protagonist's expected discounted return under dynamics parameters $\psi \sim \mathcal{D}$ as

$$J(\theta_p, \{\theta_{\text{adv}_i}\}_{i=1}^N; \{\alpha_i\}_{i=1}^N) = \mathbb{E}_{\psi \sim \mathcal{D}} \mathbb{E}_{\tau \sim (\pi_p, \{\pi_{\text{adv}_i}\}, P_\psi, \{\alpha_i\})} \left[\sum_{t=0}^{T-1} \gamma^t r(s_{i,t}, a_{i,t}^{\text{mix}}) \right] \quad (5)$$

where the trajectory τ is generated by $a_{i,t}^p \sim \pi_p(\cdot | s_{i,t})$, $a_{i,t}^{\text{adv}_i} \sim \pi_{\text{adv}_i}(\cdot | s_{i,t})$, and the environment transition P_ψ . The corresponding minimax objective can be written as

$$\max_{\theta_p} \min_{\{\theta_{\text{adv}_i}\}_{i=1}^N} J(\theta_p, \{\theta_{\text{adv}_i}\}_{i=1}^N; \{\alpha_i\}_{i=1}^N) \quad (6)$$

This adaptive adversarial mechanism introduces both intelligence and flexibility into training. The mechanism not only achieves dynamic difficulty adjustment to enhance the depth of policy evolution, but also significantly strengthens the diversity and challenge intensity of the policy learning process. As adversaries evolve and exit independently during training, the protagonist encounters a constantly evolving population of disturbances, encouraging broader and finer-grained exploration in policy space. Furthermore, training termination is no longer dictated by external iteration settings, but instead triggered by the endogenous signal that all adversaries become inactive, yielding stronger interpretability in the overall training process.

In real-world autonomous driving, policy actions are executed through a perception–planning–control pipeline and are inevitably affected by non-idealities in the control channel, such as sensor noise and state-estimation errors, actuator imperfections or partial failures (e.g., saturation or reduced authority), and communication/compute latency that introduces time misalignment between intended and executed steering/acceleration commands. Modeling the adversary as bounded perturbations in the low-level control space therefore provides a unified and physically grounded abstraction for stressing robustness under realistic execution uncertainty. Although this formulation does not explicitly generate adversarial trajectories for other traffic

participants, diversified perturbation modes can still induce interaction-relevant challenges in a closed-loop driving environment: disturbances that alter turning–deceleration coupling, throttle–brake coordination, or effectively delay/attenuate steering will change the ego vehicle's speed profile and lateral position, which in turn affects headway, time-to-collision margins, and gap acceptance during merging or car-following. As a result, the protagonist is exposed to a broader range of effective dynamic-interaction outcomes indirectly created by perturbed ego behavior, partially covering the complexity of traffic flow while keeping the adversarial mechanism controllable and reproducible.

The ACATF offers several notable benefits. First, the multi-stage progressive mechanism effectively balances exploration and stability, preventing early-stage failure due to excessive difficulty while avoiding late-stage stagnation from insufficient challenges. Second, perturbations modeling that spans from low-level physical noise to high-level policy interference achieves comprehensive coverage of multi-source uncertainties, enabling the agent to develop policies with strong generalization capabilities across diverse disturbance scenarios. Third, the mixed-action mechanism combined with adaptive disturbance regulation forms a performance-feedback-based closed-loop training system that autonomously escalates difficulty and evolves without manual intervention, thereby ensuring long-term training stability. More importantly, the framework enhances training policy diversity, game-theoretic properties, and behavioral richness through multi-adversary heterogeneous policy modeling, thereby significantly reinforcing the generalization ability of the final protagonist policy. Furthermore, as adversaries become inactive, their computational resources can be promptly released or reallocated to other active modules, improving resource efficiency and better aligning with the resource scheduling requirements of reinforcement learning platforms.

In summary, the ACATF provides a training procedure with clear structure and coherent mechanisms. By addressing training stability, policy diversity, generalization capacity, and resource efficiency, it offers a powerful paradigm for multi-agent reinforcement learning that is particularly suited for highly dynamic and complex intelligent control tasks such as autonomous driving.

4. Experiments

This section begins by outlining the overall experimental setup and environment configuration in Section 4.1. We then present a comprehensive evaluation to demonstrate the efficacy of our method: Section 4.2 benchmarks our approach against several state-of-the-art algorithms, assessing its generalization capability across diverse dynamic environments. Following this, Section 4.3 details ablation studies that verify the individual and synergistic contributions of our core components—namely, the multi-policy generation architecture, the curriculum-based adversarial training, and the diversity regularization mechanism. Section 4.4 presents a post-elimination re-evaluation experiment to verify that the protagonist retains robustness against previously defeated adversaries. Section 4.5 conducts a sensitivity analysis on key hyperparameters (e.g., the number of adversaries and the diversity coefficient) to elucidate their impact on training stability, convergence speed, and final performance, offering practical insights for algorithm deployment. Finally, Section 4.6 provides behavioral visualizations to analyze the action-level and trajectory-level characteristics of the trained policies.

4.1. Experimental setup

Our experiments are conducted on MetaDrive, a high-fidelity driving environment platform designed for autonomous driving and reinforcement learning research. Built upon the Bullet physics engine, MetaDrive supports realistic vehicle dynamics modeling and precise

collision detection while maintaining full compatibility with the OpenAI Gym interface. The platform features advanced procedural generation, enabling the automatic creation of diverse and scalable driving scenarios – ranging from lane-keeping and overtaking to complex merging and multi-vehicle interactions – which provides an efficient and flexible training environment for evaluation. MetaDrive serves as a reliable and widely adopted benchmark in academic studies for reinforcement learning, multi-agent games, and complex decision-making tasks.

To systematically evaluate the generalization of protagonist policies in complex dynamic environments, we build a multi-stage adversarial training and evaluation pipeline based on the MetaDrive platform. In our experiments, the VaryingDynamicsEnv environment of MetaDrive is used as the main platform for policy training and evaluation. This environment allows users to flexibly configure multiple key vehicle dynamics parameters, thereby replicating real-world uncertainties such as model variation, road condition changes, and component degradation. This provides controllable and diverse training conditions for robust policy learning.

4.1.1. Action space

MetaDrive adopts a continuous low-level control action space. In our setting, the action at time t is defined as $a_t = [a_t^{steer}, a_t^{acc}] \in \mathbb{R}^2$, where a_t^{steer} controls the steering command and a_t^{acc} controls longitudinal acceleration (positive for throttle and negative for braking). Following the MetaDrive Gym interface, both dimensions are bounded and we use the default normalized ranges $a_t^{steer} \in [-1, 1]$ and $a_t^{acc} \in [-1, 1]$. To ensure consistent scaling across dimensions and to avoid the diversity measure being dominated by action magnitude, we explicitly clip actions to the environment bounds and treat the bounded actions as normalized controls: $\tilde{a}_t = \text{clip}(a_t, a_{\min}, a_{\max})$, $a_{\min} = [-1, -1]$, $a_{\max} = [1, 1]$. All action-based computations in Section 3.3 ACATF Stage III (e.g., action mixing) and diversity-related statistics are performed on these normalized actions $\tilde{a}_t$.

4.1.2. State space

MetaDrive provides a vectorized observation by default. In our experiments, we use the default LidarStateObservation, where the state at time t is a normalized vector $s_t \in [0, 1]^{259}$. This state vector concatenates three components: (i) ego-vehicle state (e.g., steering/heading/velocity and relative distances to road boundaries), (ii) navigation features derived from upcoming route checkpoints (including road direction/curvature-related information), and (iii) surrounding information encoded by a LiDAR-like representation (typically a 240-beam scan within a local radius, optionally augmented with nearby-vehicle relative pose/heading). All components are normalized to $[0, 1]$ and concatenated into a single 1D vector used by the policy network.

4.1.3. Reward function

We adopt MetaDrive's default reward design, where the per-step reward is composed of three parts: a dense driving-progress reward, a dense speed reward, and a sparse termination reward. Specifically, the total reward at time step t is

$$r_t = r_t^{\text{drive}} + r_t^{\text{speed}} + r_t^{\text{term}}. \quad (7)$$

Driving-progress reward. To encourage the agent to move toward the destination along the current reference lane, we use a dense progress reward defined by the longitudinal displacement between two consecutive steps:

$$r_t^{\text{drive}} = d_t - d_{t-1}, \quad (8)$$

where d_t and d_{t-1} denote the longitudinal coordinates of the ego vehicle projected onto the current reference lane at time steps t and $t - 1$, respectively. This term provides positive reward when the vehicle makes forward progress and negative reward when it moves backward.

Speed reward. To incentivize efficient driving, we add a speed reward proportional to the current velocity:

$$r_t^{\text{speed}} = \frac{v_t}{v_{\max}}, \quad (9)$$

where v_t is the ego vehicle's speed at time t and $v_{\max}$ is the maximum allowed speed (set to 80 km/h in our experiments). This term encourages the agent to maintain a reasonably high speed when it is safe to do so.

Termination reward. At the end of an episode, all dense rewards are disabled and only a single sparse termination reward is assigned according to the termination condition:

$$r_t^{\text{term}} \in \left\{ \begin{array}{l} +\text{success_reward}, -\text{out_of_road_penalty}, \\ -\text{crash_vehicle_penalty}, -\text{crash_object_penalty}, 0 \end{array} \right\}. \quad (10)$$

Concretely, if the agent reaches the destination within the time limit, it receives a positive terminal reward `success_reward`. Otherwise, the episode terminates with a negative penalty if the vehicle goes off-road (`out_of_road_penalty`), collides with another vehicle (`crash_vehicle_penalty`), or collides with static objects (`crash_object_penalty`). The penalty is implemented as a negative reward. If the episode ends due to timeout (i.e., reaching the horizon without success or failure events), we set the terminal reward to 0 by default. Unless otherwise stated, we follow the MetaDrive default reward configuration in our experiments.

4.1.4. Evaluation protocol and difficulty settings

For each parameter configuration, multiple episodes are executed, and the results are evaluated in terms of average reward, success rate, collision rate, off-road rate, and timeout rate. To comprehensively assess policy out-of-distribution generalization, environment parameters are divided into three difficulty levels, which are further combined into six different train–test difficulty pairs:

Easy: Dynamics parameters are fixed at ideal and stable values, with no perturbations introduced.

Medium: Parameters are uniformly sampled from a moderate range, replicating a certain level of random disturbances and uncertainty.

Hard: Parameters are sampled from more extreme range that is disjoint from the previous two, incorporating extreme dynamics such as high vehicle mass, low tire friction, or nonlinear steering sensitivity.

The parameter ranges in Table 2 are chosen to be both physically meaningful and sufficiently discriminative for OOD generalization evaluation. Specifically, we follow the recommended dynamics randomization limits provided by MetaDrive's VaryingDynamicsEnv configuration (which also serves as the default randomization setup), where key vehicle dynamics parameters such as `max_engine_force`, `max_brake_force`, `wheel_friction`, `max_steering`, and `mass` are randomized within simulator-supported lower/upper bounds (Li et al., 2022). In addition, the selection of these parameters is aligned with vehicle dynamics literature (Rajamani, 2012), where vehicle mass, tire–road friction/adhesion, and steering authority are widely recognized as dominant factors governing longitudinal/lateral response, handling stability, and overall control difficulty. Consequently, we define Easy/Medium/Hard regimes by progressively expanding the dynamics shift from nominal settings to more extreme but feasible values, ensuring clear separation between difficulty levels while remaining grounded in realistic vehicle behavior. Although the parameter ranges in Table 2 are chosen to be representative for a passenger-car setting (as in MetaDrive's default configuration), our proposed training framework is not tied to any specific vehicle type or parameter interval. In principle, the method only requires bounded, configurable dynamics parameters,

Table 2

Environmental parameters for different difficulty levels.

Difficulty level	max_engine_force	max_brake_force	wheel_friction	max_steering	mass
Easy (fixed value)	2350	400	1.4	45	900
Medium (range)	[2025,2675]	[350,450]	[1.05,1.75]	[32.5,57.5]	[700,1100]
Hard (range)	[1700,2025]∪ [2675,3000]	[300,350]∪ [450,500]	[0.7,1.05]∪ [1.75,2.1]	[20,32.5]∪ [57.5,70]	[500,700]∪ [1100,1300]

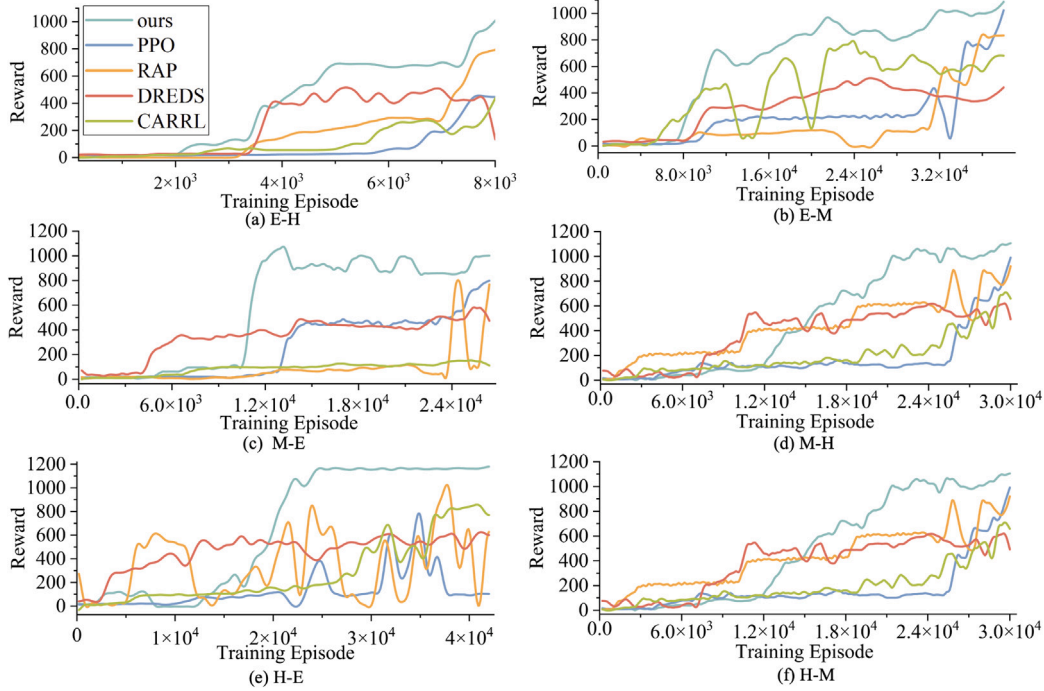


Fig. 4. Test reward versus training rounds curve for this algorithm versus different algorithms across six settings. (The legend is shown only in subfigure (a) and is identical for all subfigures (a-f)).

and the same pipeline can be applied to other vehicle classes (e.g., buses or trucks) by substituting the corresponding realistic parameter ranges and constraints.

To comprehensively evaluate the generalization ability of the proposed algorithm under varying distributional shifts, we construct six training-testing combinations based on the three disjoint difficulty levels (Easy, Medium, Hard). Each combination is denoted in the form of Train-Test, where:

- E-H: Trained on Easy and tested on Hard, this configuration quantifies extreme out-of-distribution generalization, challenging policies to operate under severe perturbations unseen during training.
- E-M: Trained on Easy and tested on Medium, this configuration evaluates upward knowledge transfer, assessing how policies adapt from simple to moderately complex scenarios.
- M-E: Trained on Medium and tested on Easy, this configuration probes backward adaptation and stability, verifying that policies retain performance when deployed in simplified environments.
- M-H: Trained on Medium and tested on Hard, this configuration examines performance escalation, measuring the policy's robustness under a direct increase in dynamic perturbation intensity.
- H-E: Trained on Hard and tested on Easy, this configuration assesses fundamental competency retention, ensuring policies do not overfit to complex conditions and remain effective in simple settings.
- H-M: Trained on Hard and tested on Medium, this configuration analyzes policy consistency and adaptability, observing its behavior when transitioning from highly complex to intermediate-difficulty tasks.

These six combinations collectively verify the algorithm's out-of-distribution generalization performance. For all experiments, the evaluation protocol is standardized: after every 200 training episodes, a 100-episode test is performed with parameters uniformly sampled from the corresponding range. The average episodic return is reported as the performance metric.

4.2. Compared with state-of-the-art methods

To comprehensively evaluate the generalization capability of our proposed method under complex perturbations and out-of-distribution dynamics, we compare it against four representative baselines: PPO (Schulman et al., 2017), RAP (Vinitsky et al., 2020), DREDS (Dai et al., 2022), and CARRL (Haoran Li et al., 2025). These methods span standard reinforcement learning, adversarial training, and domain randomization, covering a broad range of approaches to robust policy learning. PPO serves as a standard policy gradient baseline, validating our method's performance in undisturbed settings. RAP employs adversarial perturbations to enhance worst-case robustness, thereby highlighting the advantages of our multi-adversary and adaptive curriculum design. DREDS utilizes dynamic domain randomization to improve cross-distribution generalization, demonstrating our framework's superior coverage and adaptability. CARRL represents a recent advance in adversarial robust learning, confirming our comprehensive advantages in dynamic disturbance adaptation, training stability, and policy diversity.

We further ensure a fair comparison in terms of model capacity. The network architecture specification (layers $\times$ hidden dimension) and the corresponding parameter counts used to reproduce each baseline

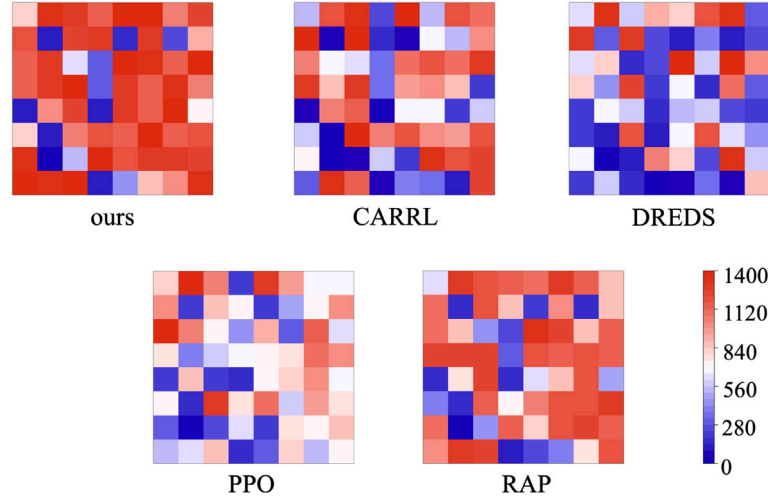


Fig. 5. Reward heatmap for different algorithms.

are reported in the Appendix. All methods adopt the same backbone depth and hidden size for the protagonist network, and any unavoidable differences in total parameters are explicitly documented and discussed. Notably, our adversary generator uses a shared-backbone multi-head design, which avoids duplicating feature extractors across adversaries; therefore, the overall parameter count of our framework does not scale linearly with the number of adversaries and can be comparable to, or even smaller than, baselines that instantiate independent networks. Overall, the comparison is fair because we control the protagonist network capacity across methods and transparently account for any remaining parameter differences, ensuring that performance improvements are not attributable to larger models.

As shown in Fig. 4, our approach consistently outperforms all baselines. Notably, the proposed framework achieves not only faster convergence and higher asymptotic performance but also effectively mitigates the training inefficiency seen in methods like RAP, where the protagonist often fails to learn when confronted with overpowering adversaries early in training. This advantage is primarily attributed to our adaptive curriculum-based adversarial training framework, which progressively escalates the adversarial challenge. By guiding the protagonist through a structured “easy-to-hard” schedule, our framework enables stable and incremental skill acquisition, ultimately fostering robust resilience to complex perturbations. Our framework integrates a shared-backbone multi-policy architecture with dual-perspective regularization to enhance adversarial diversity and perturbation coverage. This mechanism enforces behavioral distinction among policies at both local and global levels, maintaining exploration while preventing mode collapse. The synergistic design enables robust learning from heterogeneous adversarial strategies, significantly improving out-of-distribution generalization in dynamic environments.

Training under large-range perturbations (subfigures. (e)–(f)) exhibits higher volatility due to extreme dynamics (e.g., low friction, high vehicle mass). While such conditions often cause near-zero reward episodes in baseline methods – distorting value estimation and delaying convergence – our method excels through two key mechanisms: (i) an adaptive curriculum that prevents premature exposure to strong perturbations for stable skill acquisition, and (ii) covariance-based diversity regularization that maintains broad adversarial coverage to avoid policy-space collapse.

Fig. 5 presents a performance heatmap comparison between our proposed method and several mainstream baselines, including DREDS,

CARRL, RAP, and PPO, under identical training configurations. Specifically, we evaluate all algorithms on a held-out test suite consisting of 64 randomly sampled environment configurations, each representing a distinct combination of vehicle dynamics parameters. The results are visualized as an 8×8 heatmap, where each cell corresponds to the average episodic reward under a particular test configuration, aggregated over 100 independent rollouts. The color intensity reflects performance: warmer colors (red) indicate higher rewards, while cooler colors (blue) denote weaker performance. Because all algorithms are trained and evaluated under identical conditions, the heatmaps provide a direct and interpretable visualization of cross-distribution generalization gaps between our method and the baselines.

Fig. 5 demonstrates our method’s consistent superiority across all training regimes (Easy/Medium/Hard), confirming its strong generalization under dynamic conditions. While all algorithms degrade under extreme parameter combinations (e.g., low friction, high mass), our approach maintains significantly higher performance—exhibiting superior adaptability and resilience to severe distribution shifts.

We further evaluate robustness from a safety-critical perspective by reporting outcome rates in addition to return-based metrics. Specifically, under each train–test configuration, we run the trained protagonist for a fixed number of test episodes and compute four safety-relevant rates: success (reaching the destination), collision, off-road, and timeout. Table 3 summarizes these statistics (formatted as S/C/O/T, in percent) for our method and representative baselines across all six train–test settings. Overall, our approach achieves consistently higher success rates while maintaining lower collision and off-road rates in most out-of-distribution regimes, indicating that the performance gains are accompanied by safer driving behavior rather than being an artifact of reward aggregation.

To evaluate the resource cost and practical deployability of different methods during training, we conduct experiments comparing average training time and memory usage, summarized in Table 4 (Memory Usage) and Table 5 (Average Step Time), respectively. Here, average training time is defined as the average per-step time across all steps in all training episodes, rather than the average wall-clock time per episode. This definition is important because, in autonomous driving tasks, the episode length is not fixed: trajectories may terminate early due to collisions, off-road events, or other termination conditions, which makes episode-level time comparisons strongly confounded by varying episode lengths. In contrast, average step time provides a more stable and comparable measure of the computational cost of one policy

Table 3

Safety metrics comparison (Success(S)/Collision(C)/Off-road(O)/Timeout(T); rates in percent).

Method	E-M	E-H	M-E	M-H	H-E	H-M
Ours	82/5/13/0	72/12/15/1	91/8/0/1	77/5/3/15	93/3/1/3	91/4/3/2
RAP	68/8/24/0	52/7/12/29	75/6/19/0	63/6/15/16	90/7/3/0	85/9/2/4
PPO	70/4/25/1	29/12/58/1	70/10/20/0	40/12/30/18	78/7/11/4	71/9/19/1
CARRL	65/13/22/0	54/21/25/0	63/16/21/0	45/23/32/0	76/11/11/2	54/6/18/22
DREDS	45/21/34/0	56/4/37/3	61/15/25/0	47/14/39/0	58/10/24/8	61/6/32/0

Table 4

Memory usage comparison (MB) across train-test scenarios.

Scenario	Ours	RAP	PPO	CARRL	DREDS
E-M	635.202	715.232	612.890	665.748	623.482
E-H	626.802	703.260	685.079	639.625	692.346
M-E	679.298	694.725	622.939	772.498	621.954
M-H	607.280	708.693	634.702	689.437	618.652
H-E	610.647	672.254	613.320	736.953	637.820
H-M	620.622	735.538	650.575	689.437	607.152

Table 5

Average step time (ms) comparison across train-test scenarios.

Scenario	Ours	RAP	PPO	CARRL	DREDS
E-M	1.478	2.489	1.622	4.692	7.848
E-H	1.511	2.267	1.222	4.158	7.455
M-E	1.329	2.057	1.467	4.769	6.944
M-H	1.243	2.572	1.589	4.642	7.190
H-E	1.511	2.550	1.441	4.548	7.335
H-M	1.544	2.784	1.524	4.440	7.144

forward pass, one environment interaction, and the associated update operations. It is also worth noting that although the per-step time differences in the tables may appear small, the full training process typically involves on the order of tens of millions of environment steps, so even minor per-step gaps can accumulate into a noticeable difference in total training time.

In addition, our training protocol follows a one-to-one interaction setting: at any time, the protagonist interacts with only one adversary policy (one head) and performs the corresponding updates, rather than interacting with multiple adversaries in parallel. This sequential interaction-and-update pattern is more suitable for CPU execution; therefore, we run all methods on CPU and report memory usage under a consistent hardware setting to ensure fair and reproducible comparisons. Overall, our approach avoids redundant computation and storage by using a shared-backbone multi-head design (instead of duplicating a full network per adversary), and activates only a single adversary head at a time in the adversarial stage. Together with curriculum-based intensity scheduling and diversity regularization, the proposed framework achieves stronger robustness and cross-distribution generalization without a substantial increase in per-step computational cost, resulting in a more favorable trade-off between performance and resource consumption.

4.3. Ablation experiment

We conduct a progressive ablation study that builds the proposed framework in a *stacking* manner, so that the contribution of each component can be isolated and the synergy among components can be directly observed. Starting from a plain baseline with no proposed modules, we progressively add one component, then two components, and finally all three components, and evaluate each variant under six train-test configurations (five repetitions each). Results are reported in two complementary tables: [Table 6](#) summarizes episodic return (mean $\pm$ std), while [Table 7](#) reports safety-relevant outcome rates from the same evaluation episodes (success/collision / off-road/timeout; all in percent). This joint reporting ensures that improvements in return

Table 6Progressive ablation study (mean reward $\pm$ std, 5 runs).

Variant	E-M	E-H	M-E
single net (0adv)	348.61 $\pm$ 27.69	327.79 $\pm$ 25.31	838.30 $\pm$ 26.29
single net (5adv)	657.12 $\pm$ 26.59	779.30 $\pm$ 28.97	929.95 $\pm$ 25.73
MPG-Arch (5adv)	826.84 $\pm$ 27.91	988.79 $\pm$ 26.89	956.01 $\pm$ 28.55
single net +DPPER (5adv)	808.81 $\pm$ 28.17	767.57 $\pm$ 22.66	1052.17 $\pm$ 25.17
single net +ACATF (5adv)	819.86 $\pm$ 22.38	756.89 $\pm$ 25.52	1022.84 $\pm$ 21.54
MPG-Arch +ACATF (5adv)	853.47 $\pm$ 27.98	990.43 $\pm$ 29.33	1096.13 $\pm$ 26.38
MPG-Arch +DPPER (5adv)	931.45 $\pm$ 26.91	963.92 $\pm$ 23.25	1098.48 $\pm$ 21.16
single net +DPPER +ACATF (5adv)	1043.70 $\pm$ 22.72	899.46 $\pm$ 20.33	1176.93 $\pm$ 25.87
Complete Model	1149.62 $\pm$ 27.45	1008.06 $\pm$ 26.23	1178.97 $\pm$ 24.35

Variant	M-H	H-E	H-M
single net (0adv)	584.79 $\pm$ 27.34	903.66 $\pm$ 29.02	557.62 $\pm$ 29.47
single net (5adv)	765.61 $\pm$ 26.11	1028.73 $\pm$ 27.28	844.51 $\pm$ 29.54
MPG-Arch (5adv)	800.18 $\pm$ 25.18	1183.28 $\pm$ 29.32	981.68 $\pm$ 27.06
single net +DPPER (5adv)	911.34 $\pm$ 26.75	1055.73 $\pm$ 28.01	999.76 $\pm$ 27.22
single net +ACATF (5adv)	854.55 $\pm$ 23.07	1042.50 $\pm$ 28.18	990.22 $\pm$ 25.83
MPG-Arch +ACATF (5adv)	906.79 $\pm$ 28.71	1231.42 $\pm$ 25.94	1096.68 $\pm$ 29.15
MPG-Arch +DPPER (5adv)	874.95 $\pm$ 22.71	1168.45 $\pm$ 20.87	1109.51 $\pm$ 21.15
single net +DPPER +ACATF (5adv)	905.70 $\pm$ 28.99	1195.60 $\pm$ 27.32	1120.82 $\pm$ 25.07
Complete Model	940.28 $\pm$ 25.65	1253.68 $\pm$ 26.32	1149.01 $\pm$ 26.19

correspond to safer driving behavior rather than being masked by reward aggregation.

To make the progression explicit, variants are organized by the number of proposed components included (MPG, DPP, ACA in the tables). **(0 component)** *single net (0adv)* removes adversarial training entirely and serves as the non-adversarial baseline; *single net (5adv)* introduces adversarial training with multiple adversaries but without MPG-Arch, DPPER, or ACATF. **(1 component)** We then add exactly one module at a time: adding only the curriculum yields *single net +ACATF (5adv)*; adding only diversity yields *single net +DPPER (5adv)*; adding only the multi-policy generator yields *MPG-Arch (5adv)* (multi-head adversaries with a shared backbone, without DPPER/ACATF). **(2 components)** Next, we test pairwise combinations: *MPG-Arch+ACATF (5adv)* removes DPPER while keeping MPG-Arch and curriculum, and *MPG-Arch+DPPER (5adv)* removes ACATF while keeping MPG-Arch and diversity. In addition, *single net +DPPER+ACATF (5adv)* keeps both training components but replaces MPG-Arch with independent adversary networks, isolating the architectural effect of a shared-backbone multi-head generator. **(3 components)** Finally, *Complete Model* includes all proposed modules (MPG-Arch + DPPER + ACATF) with 5 adversaries.

As more components are stacked, both returns ([Table 6](#)) and safety outcomes ([Table 7](#)) improve consistently across the six train-test settings, and the **Complete Model** achieves the strongest overall performance: it yields the highest returns while also attaining the best safety

Table 7

Safety metrics comparison (Success(S)/Collision(C)/Off-road(O)/Timeout(T); rates in percent).

Variant	E-M	E-H	M-E
single net (0adv)	23/22/55/0	17/47/36/0	56/4/37/3
single net (5adv)	41/13/46/0	37/7/19/37	65/10/24/1
MPG-Arch (5adv)	57/12/31/0	70/9/20/1	65/8/25/2
single net +DPPER (5adv)	53/13/31/3	49/6/16/29	73/12/1/14
single net +ACATF (5adv)	54/6/18/22	51/10/38/1	73/6/21/0
MPG-Arch +ACATF (5adv)	58/20/22/0	70/10/20/0	77/9/11/3
MPG-Arch +DPPER (5adv)	65/10/24/1	68/13/19/0	79/5/16/0
single net +DPPER+ACATF (5adv)	73/4/23/0	61/6/32/1	86/11/1/2
Complete Model	82/5/13/0	72/12/15/1	91/8/0/1

Variant	M-H	H-E	H-M
single net (0adv)	36/19/43/2	63/16/21/0	34/24/42/0
single net (5adv)	49/6/16/29	73/9/18/0	57/4/39/0
MPG-Arch (5adv)	55/20/25/0	85/9/2/4	69/8/22/1
single net +DPPER (5adv)	63/7/29/1	76/8/16/0	70/9/21/0
single net +ACATF (5adv)	58/20/22/0	75/8/17/0	70/10/20/0
MPG-Arch +ACATF (5adv)	63/16/21/0	90/9/1/0	78/12/10/0
MPG-Arch +DPPER (5adv)	60/14/25/1	83/5/2/10	78/9/8/5
single net +DPPER+ACATF (5adv)	71/12/15/2	86/6/8/0	79/6/15/0
Complete Model	77/5/3/15	93/3/1/3	91/4/3/2

Table 8

Post-training adversarial re-evaluation against saved adversaries.

Variant	E-M	E-H	M-E
Protagonist (no adv)	1149.62 ± 27.45	1008.06 ± 26.23	1178.97 ± 24.35
Protagonist + adv1	1023.45 ± 24.88	924.53 ± 22.91	1092.83 ± 23.16
Protagonist + adv2	992.65 ± 22.37	947.50 ± 26.08	1107.96 ± 24.71
Protagonist + adv3	1014.75 ± 23.92	973.83 ± 23.44	1094.55 ± 21.95
Protagonist + adv4	1007.24 ± 25.16	946.35 ± 21.63	1083.46 ± 26.22
Protagonist + adv5	1025.64 ± 21.74	953.34 ± 25.27	1075.93 ± 22.84
Best baseline	835.40 ± 28.03	883.76 ± 27.19	924.61 ± 29.04

Variant	M-H	H-E	H-M
Protagonist (no adv)	940.28 ± 25.65	1253.68 ± 26.32	1149.01 ± 26.19
Protagonist + adv1	934.67 ± 24.09	1141.15 ± 22.48	1089.53 ± 22.75
Protagonist + adv2	863.50 ± 20.77	1116.11 ± 24.96	1084.78 ± 25.09
Protagonist + adv3	916.19 ± 25.68	1193.70 ± 23.07	1104.53 ± 23.66
Protagonist + adv4	903.14 ± 23.35	1159.69 ± 25.41	1123.98 ± 21.34
Protagonist + adv5	910.10 ± 21.52	1168.32 ± 21.89	1118.60 ± 26.73
Best baseline	895.83 ± 27.61	1012.39 ± 28.54	991.23 ± 29.17

profile (higher success rates with reduced collision, off-road, and timeout rates). Comparing **1-component** variants against **0-component** baselines shows that each proposed module provides a measurable gain: ACATF improves stability under increasing disturbance intensity through performance-triggered scheduling, DPPER promotes heterogeneous adversarial behaviors and mitigates mode collapse, and MPG-Arch improves behavioral coverage via efficient multi-head generation with shared representation. Moving from **1-component** to **2-component** variants yields further gains, indicating that training strategy (ACATF), diversity shaping (DPPER), and architecture (MPG-Arch) are complementary rather than redundant. Finally, stacking all three components produces the best results.

4.4. Post-elimination adversary re-evaluation

To examine whether the protagonist truly becomes robust to the adversaries it has already “defeated” during Stage III, we perform a post-elimination re-evaluation experiment. During adversarial training, whenever an adversary head is deactivated (i.e., eliminated by the curriculum rule), we checkpoint and save the corresponding adversary model. After Stage III terminates, we reload the final protagonist and individually re-test it against each saved adversary head under the same evaluation protocol.

For each train-test configuration, we report the protagonist’s standalone performance (no adversary) and its performance when mixed

with a saved adversary adv_i via $a^{\text{mix}} = \alpha a^{\text{adv}_i} + (1 - \alpha) a^p$, using a strong mixing coefficient $\alpha = 0.4$. This post-hoc test evaluates whether the final protagonist sustains high returns even when previously eliminated adversaries are reintroduced with stronger influence.

As shown in Table 8, the protagonist remains robust against all saved adversaries: *Protagonist+adv_i* returns stay close to the standalone *Protagonist (no adv)* scores across configurations, and consistently surpass the strongest baseline (RAP) even at $\alpha = 0.4$. For instance, in H-E, the protagonist scores 1253.68 alone and 1193.70 under *Protagonist+adv₃*, confirming that adversary elimination reflects genuine robustness rather than overfitting.

4.5. Impact of key hyperparameters

To systematically evaluate the influence of key hyperparameters on generalization, we design two sets of comparative experiments. First, we vary the number of adversaries $\{1, 3, 5, 7, 9\}$ while fixing the diversity coefficient to assess multi-policy network scaling effects. Second, we adjust the diversity parameter $h \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$ while maintaining a fixed adversary count to evaluate diversity constraint influences. All experiments are conducted across six train-test environment combinations, with performance compared through training curves and evaluation returns to quantify effects on stability, convergence speed, and performance ceiling.

Fig. 6 demonstrates that the number of adversaries significantly affects policy performance across all environments. With 1–3 adversaries, the protagonist quickly masters their limited perturbation modes, leading to insufficient challenge and poor out-of-distribution generalization. Five adversaries achieve the optimal balance—providing enough diversity to prevent overfitting while maintaining training stability, yielding consistently superior returns. However, increasing to 7–9 adversaries introduces excessive challenge, causing training oscillations and unstable final performance. These findings highlight that moderate adversary counts best balance diversity and stability.

Fig. 7 illustrates the cumulative reward trend of the protagonist policy during training under different adversarial diversity coefficients. Overall, moderate diversity constraints substantially enhance training efficiency and the policy’s final performance, whereas both excessively low and high diversity coefficients present shortcomings. Specific analysis reveals distinct patterns across diversity coefficients. With weak constraints ($h = 0.1$ – 0.2), limited adversary diversity causes policy overfitting to fixed patterns, manifesting as slow convergence and suboptimal asymptotic performance. Conversely, strong constraints ($h = 0.4$ – 0.5) yield overspread perturbations that, while enhancing exploration, introduce excessive gradient noise—resulting in training volatility and convergence delays. The moderate setting ($h = 0.3$) optimally balances these effects: maintaining sufficient policy differentiation to escape local optima while avoiding destabilizing variance, thus achieving both fastest convergence and highest final returns across testing scenarios.

Fig. 8 compares diversity metrics (all with $h = 0.3$), including our hierarchical covariance method alongside cosine similarity, KL divergence, within-state only (micro), across-state only (macro), and the Vendi Score baseline (computed on normalized actions using a standard RBF kernel). Our method outperforms all alternatives—especially in challenging regimes. Its advantages are threefold: First, while pairwise metrics (KL divergence, cosine similarity) only capture local relationships, and the Vendi Score depends on a manually specified kernel function that is sensitive to scale, our covariance determinant measures global policy distribution coverage with a stable, differentiable training signal, preventing dimensional collapse. Second, covariance determinants provide a clear geometric interpretation of distribution volume and avoid the $O(N^2)$ kernel matrix construction required by the Vendi Score, making them more computationally favorable for repeated evaluation during training. Finally, our hierarchical approach

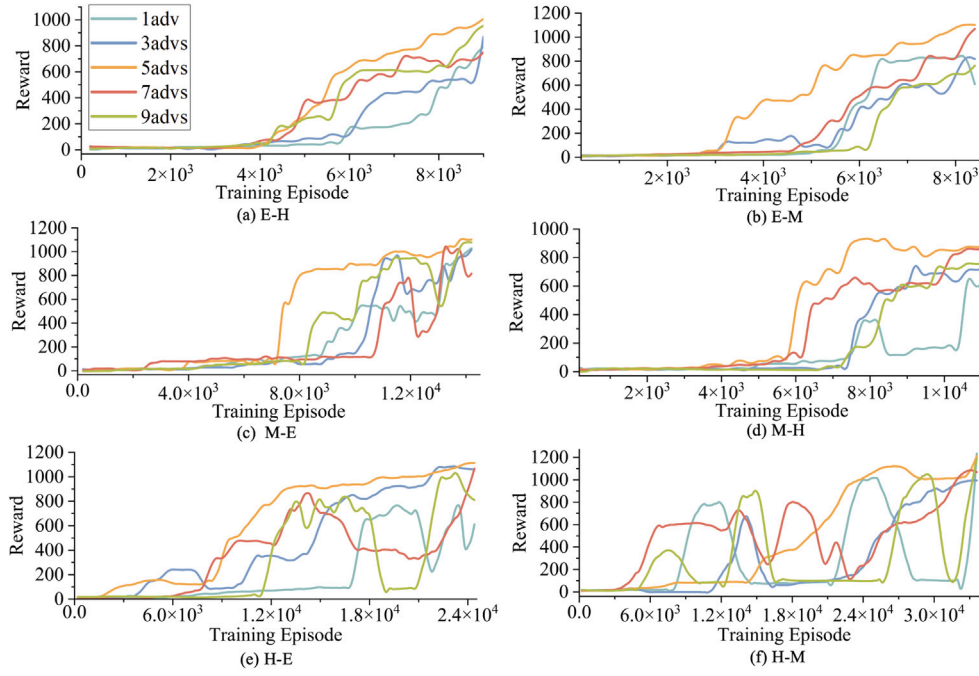


Fig. 6. Test reward-training round count curves across different adversary numbers in six settings. (The legend is shown only in subfigure (a) and is identical for all subfigures (a–f)).

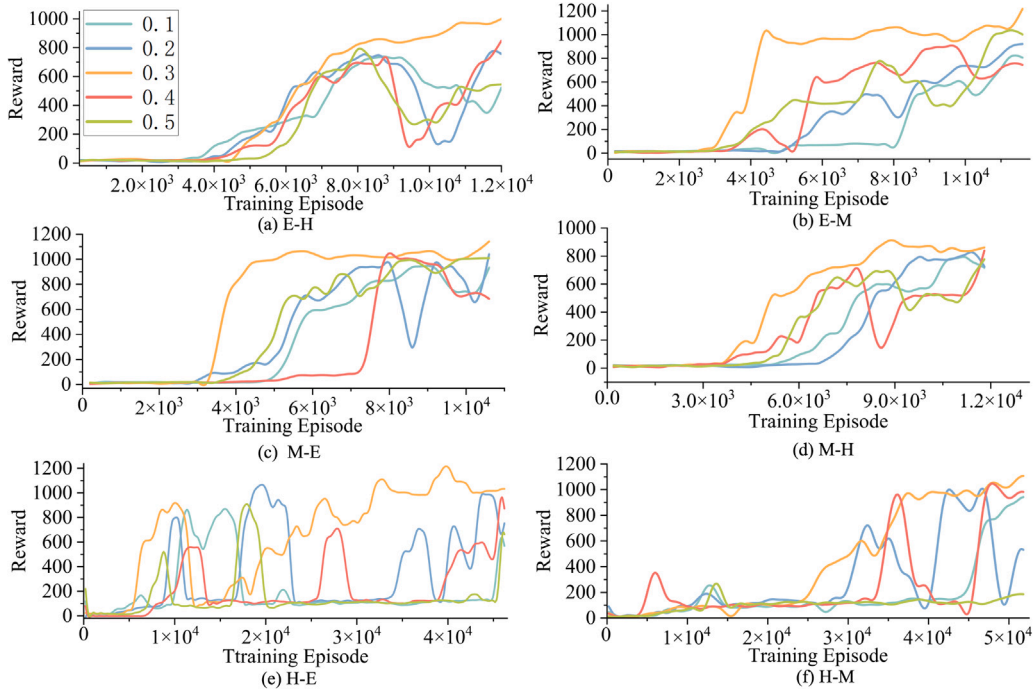


Fig. 7. Test reward-training round count curves across diversity coefficients h in six settings. (The legend is shown only in subfigure (a) and is identical for all subfigures (a–f)).

balances within-state variation with cross-state distributional coverage, preventing stage-specific imbalance seen in single-perspective metrics.

In Section 3.3, the predefined threshold τ controls the performance-triggered adjustment of adversarial intensity (i.e., when to strengthen or relax the mixing schedule), and thus directly affects the pacing of Stage-III curriculum. As shown in Table 9, we conduct a parameter sensitivity study by training the full framework with three representative threshold values, $\tau \in \{800, 1000, 1200\}$, while keeping all other settings fixed. The results show that $\tau = 1200$ yields the most

consistent improvements under challenging regimes, whereas smaller thresholds tend to degrade robustness. Intuitively, when τ is set too low (e.g., 800 or 1000), the system may adjust the adversarial intensity prematurely: the protagonist is deemed “sufficiently strong” early on, leading to earlier deactivation of adversaries or insufficient exposure to harder disturbances, which encourages mode-specific adaptation rather than broad generalization. In contrast, $\tau = 1200$ enforces a stricter competence requirement before progressing the curriculum, maintaining adversarial pressure for longer and promoting sustained

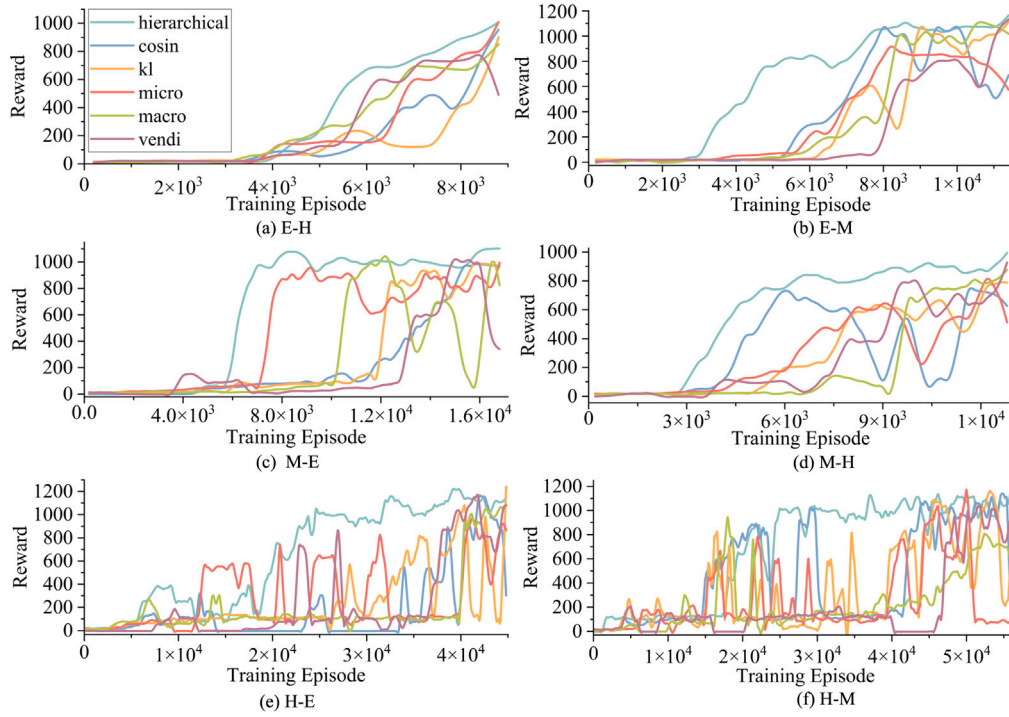


Fig. 8. Test reward versus training rounds curves across different diversity metric parameters in six settings. (The legend is shown only in subfigure (a) and is identical for all subfigures (a–f)).

exploration. As a result, the protagonist experiences a richer and more challenging disturbance population during training, which better aligns with the objective of learning robust behaviors across dynamics shifts, explaining why $\tau = 1200$ achieves the best overall performance in our experiments.

Guidelines for selecting τ . In practice, τ does not need to be treated as an absolute, environment-specific constant. A simple and transferable guideline is to use a *capability reference* computed from the end of Stage II: we take the moving-average return over the last several evaluation windows (e.g., the Stage II running average over K episodes) as R_{ref} , and set τ to roughly 80–90% of this reference, i.e., $\tau = \rho \cdot R_{\text{ref}}$ with $\rho \in [0.8, 0.9]$. This makes the threshold automatically scale with the reward magnitude of the environment and reduces manual tuning when transferring to a new task. When reward scales differ substantially across tasks (e.g., the overall return magnitude changes due to different reward shaping or episode horizons), a *normalized threshold* can be used instead, such as defining the trigger based on a percentile of the running-average return (e.g., requiring $\bar{R}^{(K)}$ to exceed the p th percentile of its recent history), which further improves robustness to reward rescaling.

We also tested a larger threshold (e.g., $\tau = 1400$), but it made the curriculum overly strict: the protagonist rarely satisfied the transition criterion, causing training to remain stuck in earlier stages and preventing the full Stage-III schedule from being effectively executed; therefore, we omit these results for clarity. Qualitatively, this “stuck” phenomenon arises because an excessively high threshold makes stage-advancement conditions extremely hard to satisfy, so the training spends a prolonged period in low-intensity adversarial interaction and fails to reach the regime where higher-strength disturbances are sufficiently encountered and learned against.

4.6. Behavioral analysis and visualization

To better understand how the proposed framework improves robustness beyond aggregate metrics (e.g., reward and success rate), we

Table 9

Sensitivity study of the reward threshold τ (mean reward $\pm$ std, 5 runs).

Scenario	$\tau = 800$	$\tau = 1000$	$\tau = 1200$
E–M	866.43 $\pm$ 20.60	966.05 $\pm$ 21.15	1149.62 $\pm$ 27.45
E–H	766.69 $\pm$ 23.98	843.79 $\pm$ 26.13	1008.06 $\pm$ 26.23
M–E	873.48 $\pm$ 21.01	984.50 $\pm$ 28.75	1178.97 $\pm$ 24.35
M–H	860.40 $\pm$ 21.28	906.91 $\pm$ 23.18	940.28 $\pm$ 25.65
H–E	1036.46 $\pm$ 23.73	1112.36 $\pm$ 24.16	1253.68 $\pm$ 26.32
H–M	911.76 $\pm$ 22.22	1077.68 $\pm$ 22.68	1149.01 $\pm$ 26.19

conduct a set of behavioral visualizations that reveal the action-level and trajectory-level characteristics of the trained policies.

Figs. 9 and 10 visualize per-step action samples collected from rollouts under the same evaluation setting. Fig. 9 focuses on the protagonist-centered view: we plot the protagonist’s action distribution together with five mixed-action distributions (Mix₁–Mix₅), where each mixed action is obtained by linearly mixing the protagonist action with one adversary head under the same mixing rule. Each point in the scatter plot corresponds to one executed action sample in the 2D action space (steering/acceleration). These plots show that action mixing perturbs the protagonist’s control while largely preserving its overall action regime, confirming that the mixed actions remain close to the protagonist behavior with only bounded deviations. To further verify this empirically, we measure the directional alignment between $a_{i,t}^p$ and $a_{i,t}^{\text{mix}_j}$ across all Stage-III training steps: the overall direction-consistency ratio is **99.495%** (per-head: adv₁: 99.798%, adv₂: 99.521%, adv₃: 99.641%, adv₄: 98.786%, adv₅: 99.643%), confirming that linear mixing preserves the protagonist’s action intent in the vast majority of cases and does not degenerate into semantic cancellation. It is worth emphasizing that the mixed action is used only during adversarial training to improve robustness; during testing and intended deployment, the vehicle is controlled solely by the protagonist policy, so such mixed-action inconsistency does not directly arise in the final decision-making stage.

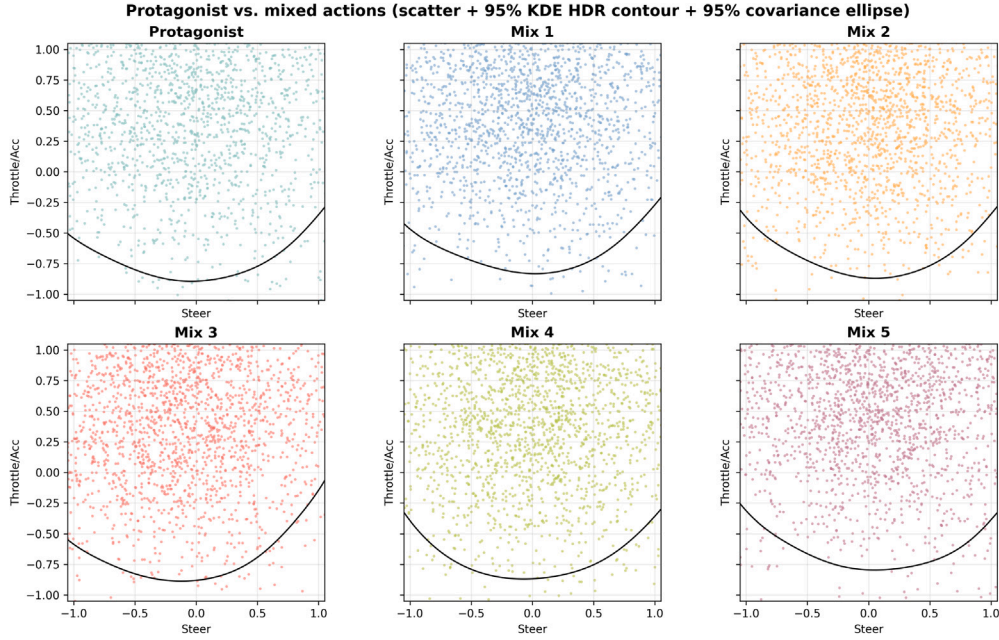


Fig. 9. The protagonist's action distribution together with five mixed-action distributions (Mix₁–Mix₅).

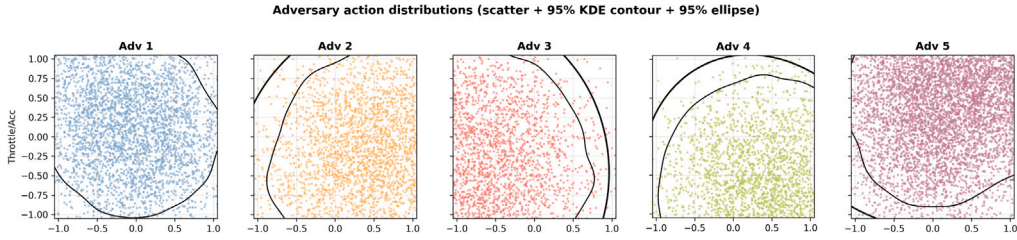


Fig. 10. The action distributions produced by the five adversary heads (Adv₁–Adv₅).

Fig. 10 presents the adversary-side view by plotting the action distributions produced by the five adversary heads (Adv₁–Adv₅). This figure is used to verify whether different heads induce heterogeneous control patterns, rather than merely changing action magnitude. To highlight the *coupling* between action dimensions, we overlay a 95% KDE HDR contour and a 95% confidence ellipse for each distribution. The ellipse is estimated from the empirical covariance of the sampled actions, and its orientation (i.e., the tilt of the major axis) reflects the dominant correlation direction of the policy's action outputs. Different tilt directions therefore indicate different behavioral tendencies (e.g., whether larger steering is associated with reduced acceleration, or whether longitudinal and lateral control are decoupled). Together, the KDE contour and the confidence ellipse serve as compact summaries of each policy's control “style” and make cross-head behavioral differences directly observable.

Finally, we report four time-series signals of the trained protagonist over representative episodes, with the horizontal axis being the time step: speed, longitudinal acceleration, steering command, and throttle/brake command (Fig. 11). These micro-level curves complement the macro statistics (reward and safety rates) by revealing how robustness is achieved in the control process: the policy maintains stable speed evolution with bounded acceleration, indicating consistent and safe longitudinal control rather than opportunistic reward chasing. Moreover, both the lateral (steering) and longitudinal (throttle/brake) control commands evolve smoothly without abrupt oscillations or high-frequency switching, suggesting that the learned robustness comes from stable, physically plausible control behaviors under perturbations. Together, these visualizations indicate that the proposed multi-head

adversarial training improves not only aggregate performance but also the behavioral reliability of driving policies.

5. Conclusion

This study addresses core challenges in adversarial reinforcement learning for autonomous driving – policy overfitting, training instability, and limited adversarial diversity – through an integrated framework combining shared-backbone multi-policy generation, hierarchical covariance-based diversity regularization, and adaptive curriculum learning, which together improve behavioral coverage, prevent mode collapse, and stabilize policy acquisition across difficulty stages. Extensive experiments in high-fidelity driving environments demonstrate consistent superiority over state-of-the-art baselines (PPO, RAP, DREDS, CARRL), with reward improvements of 22.75%, 7.36%, 50.06%, and 27.35%, respectively. Ablation studies confirm the necessity of each component, and the framework maintains reliable performance under complex dynamics and strong perturbations, confirming its generalization capability and demonstrating promise for real-world autonomous driving applications, although the current validation remains limited to high-fidelity simulation.

Several directions remain open for future investigation. First, the adversarial setting could be extended to model environment-level traffic participants (e.g., aggressive cut-ins or sudden braking), better reflecting realistic multi-agent driving scenarios. Second, dynamic head allocation for the multi-policy generator – adaptively pruning or expanding adversary heads based on diversity and utility – would improve

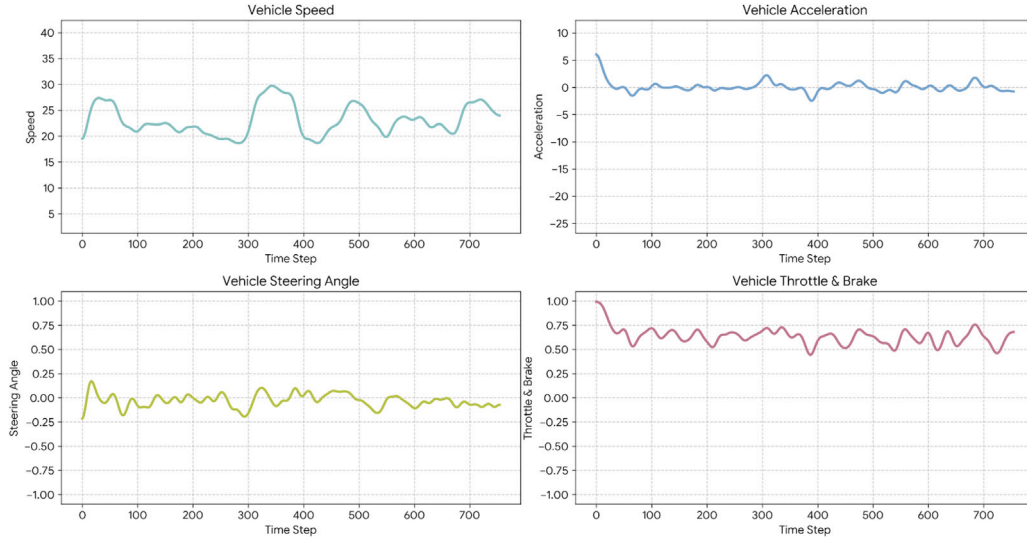


Fig. 11. Time-Domain Analysis of Vehicle Dynamics and Interaction Signals.

scalability. Third, replacing the fixed reward threshold τ with an adaptive schedule driven by recent performance statistics or safety-oriented signals (success, collision, and off-road rates) could enable more responsive curriculum progression across tasks with varying reward scales. In addition, the current framework does not yet resolve several key sim-to-real challenges, including sensor noise and perception errors (e.g., LiDAR/camera uncertainty), communication/computation delays, and the modeling gap between MetaDrive physics and real vehicle dynamics. Future work will therefore consider extended domain randomization over sensor-noise parameters, joint validation with real-world datasets such as nuScenes or Waymo, and preliminary transfer experiments on small-scale real vehicle platforms. We also plan to perform a dedicated safety audit of the rare direction-inconsistent moments observed during adversarial mixing, in order to more precisely quantify their correlation with collisions, off-road events, and other safety-critical indicators.

CRedit authorship contribution statement

Xiaofeng Wang: Writing – review & editing, Supervision, Resources, Funding acquisition. **Shaopu Wang:** Writing – original draft, Methodology, Investigation, Formal analysis, Data curation. **Jipeng Zhang:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Yue Ma:** Resources, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Network configurations (baselines)

All methods share the same backbone depth and hidden size. The protagonist uses a 3-layer MLP (hidden dim 64). Our adversary adopts a shared-backbone multi-head design with $N = 5$ heads, keeping total parameters at 21,775—far fewer than RAP’s 5 independent networks (124,717). DREDS uses a single network with augmentation layers (83,138), while CARRL and PPO use standard actor–critics comparable to our protagonist (41,795). All methods use Adam with identical learning rate and discount factor, and on-policy rollouts (up to 1000 steps per episode). Our method applies ACATF’s performance-triggered early stopping (budget: 10,000+20,000+60,000 epochs across three stages); baselines train for a fixed 90,000 episodes. Full architectural details and hyperparameters are summarized in Table A.10.

Appendix B. Hyperparameter settings and optimization details

B.1. Network architectures

We use MetaDrive’s default vectorized observation $s_t \in [0, 1]^{259}$ and a continuous action $a_t \in \mathbb{R}^2$. The protagonist adopts a standard MLP actor–critic with two hidden layers (Tanh): Actor $259 \rightarrow 64 \rightarrow 64 \rightarrow 2$ and Critic $259 \rightarrow 64 \rightarrow 64 \rightarrow 1$. The adversary follows a shared-backbone multi-head actor–critic design with $N = 5$ independent heads.

B.2. Core hyperparameters

Unless otherwise stated, we use Adam with learning rate 3×10^{-4} , discount factor $\gamma = 0.99$, PPO clipping $\epsilon = 0.2$, and entropy coefficient 0.01. We set $K_{\text{epochs}} = 20$ for the protagonist and $K_{\text{epochs}} = 10$ for adversaries. Training is on-policy with a dynamic batch (one full episode rollout, up to 1000 steps) per update.

B.3. Three-stage optimization procedure

Stage I pretrains the protagonist on fixed nominal dynamics for up to 10,000 epochs. Stage II performs domain-randomized training for up to 20,000 epochs, where dynamics parameters are uniformly sampled from the Medium ranges (Table 2) at each epoch reset. Stage III conducts adversarial training with $N = 5$ adversaries and action mixing $a^{\text{mix}} = \alpha_i a^{\text{adv}_i} + (1 - \alpha_i) a^p$, using $R_p = R_{\text{env}}$ and $R_{\text{adv}_i} = -R_{\text{env}}$. Every 20 epochs, α_i is adaptively adjusted based on recent performance

Table A.10

Network architecture and key hyperparameters for our method and baselines.

Item	Ours	PPO	RAP	DREDS	CARRL
Network architecture (layers $\times$ hidden dim)	3×64	3×64	3×64	3×64	3×64
Parameter count (protagonist)	41,795	41,795	41,795	83,138	41,795
Parameter count (adversary)	21,775	–	124,717	–	–
Learning rate	3×10^{-4}	3×10^{-4}	3×10^{-4}	3×10^{-4}	3×10^{-4}
Discount factor γ	0.99	0.99	0.99	0.99	0.99
Max training episodes	90,000	90,000	90,000	90,000	90,000

Table C.11

Training time comparison (hours) across train–test scenarios.

Scenario	Ours	RAP	PPO	CARRL	DREDS
E–M	1.23	10.23	8.12	15.42	37.60
E–H	0.92	9.10	9.51	16.43	35.93
M–E	2.34	11.43	7.71	19.36	30.25
M–H	3.15	10.36	6.23	20.81	33.11
H–E	6.13	11.73	9.75	26.79	38.15
H–M	7.42	12.91	9.36	25.43	31.45

and the adversary is deactivated if $\alpha_i \geq 0.5$; training terminates when all adversaries are inactive or the epoch count reaches 60,000.

B.4. Evaluation

We evaluate every 200 training epochs by testing the protagonist (without adversaries) for 100 episodes under the corresponding test difficulty settings, and report average return and safety metrics (success, collision, off-road, and timeout rates).

Appendix C. Training time comparison

Table C.11 reports the wall-clock training time (in hours) for each method across all train–test scenarios. Our proposed method consistently requires far less total training time than all baselines, particularly against CARRL and DREDS, which incur the highest overhead due to their domain-randomization and ensemble-based designs. However, it is crucial to note that this macro-level time comparison does not reflect an equal learning budget. The baselines were trained for a fixed 90,000 episodes regardless of convergence. In contrast, the significant reduction in total training time for ACATF is primarily driven by its performance-triggered early stopping mechanism—specifically, adversaries are progressively deactivated once defeated, and training terminates entirely when all adversaries are inactive. Therefore, the total time difference in Table C.11 primarily highlights the overall sample efficiency of the curriculum design.

Data availability

Data will be made available on request.

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